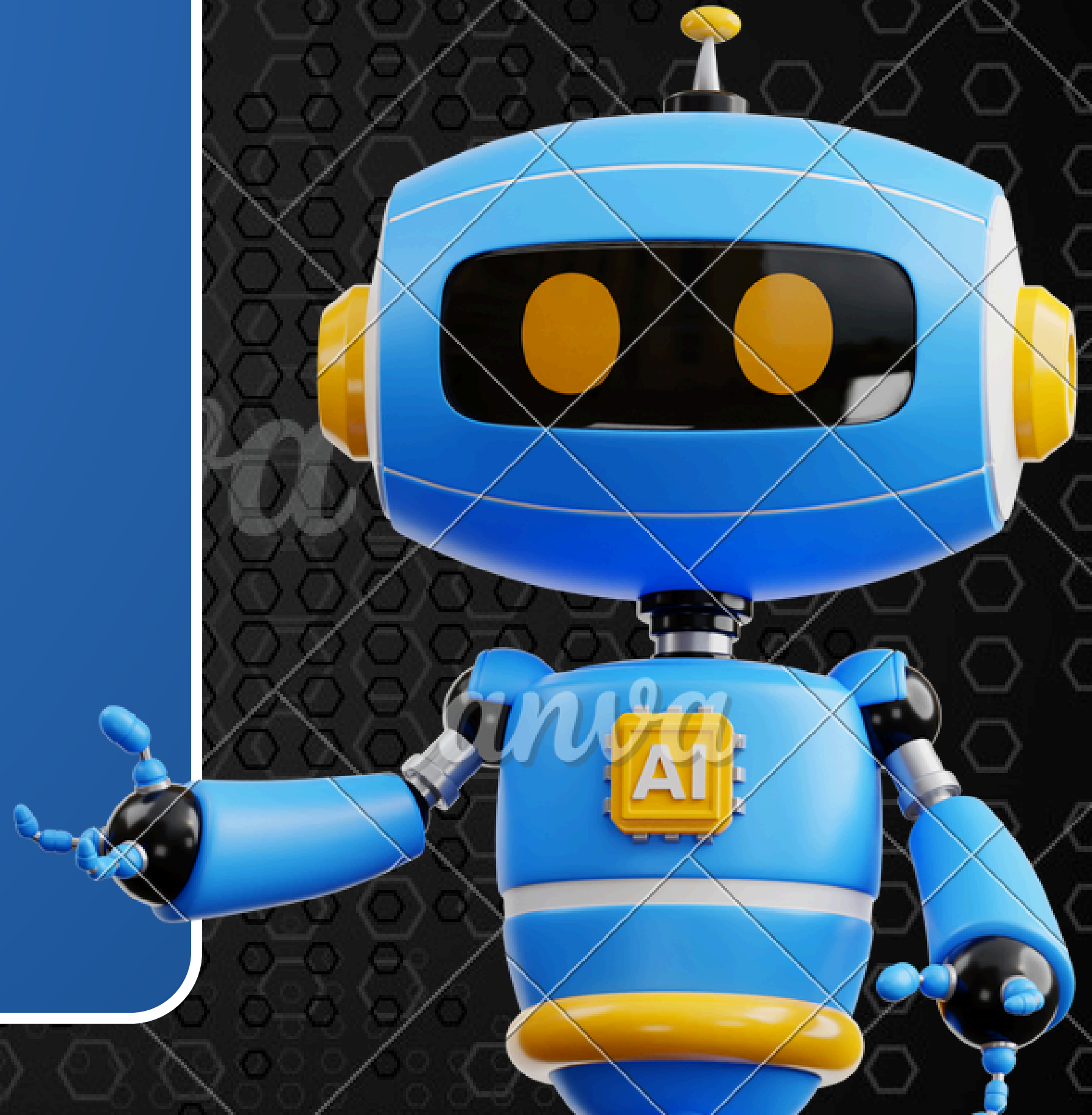
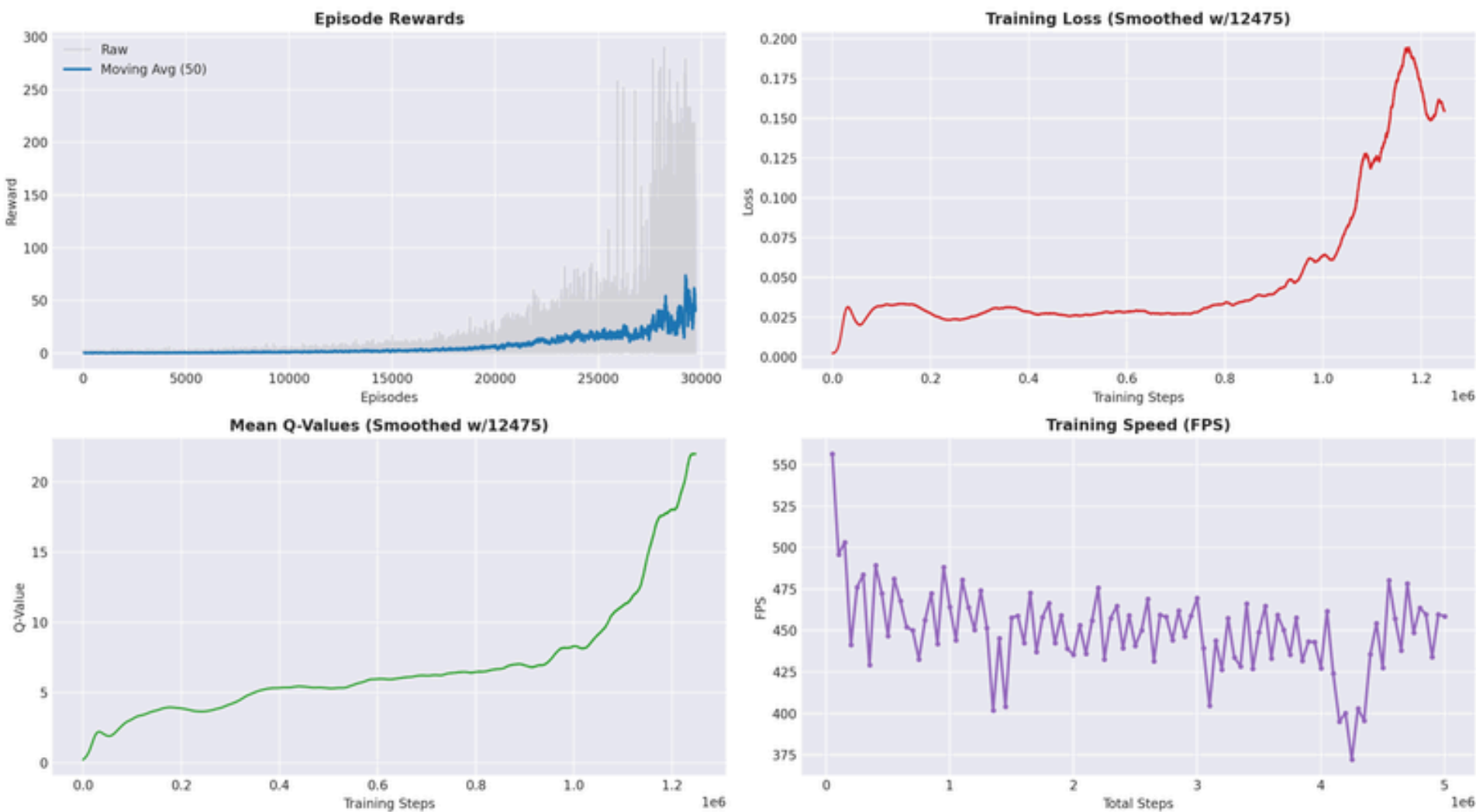


Atari Agents

Rezultate

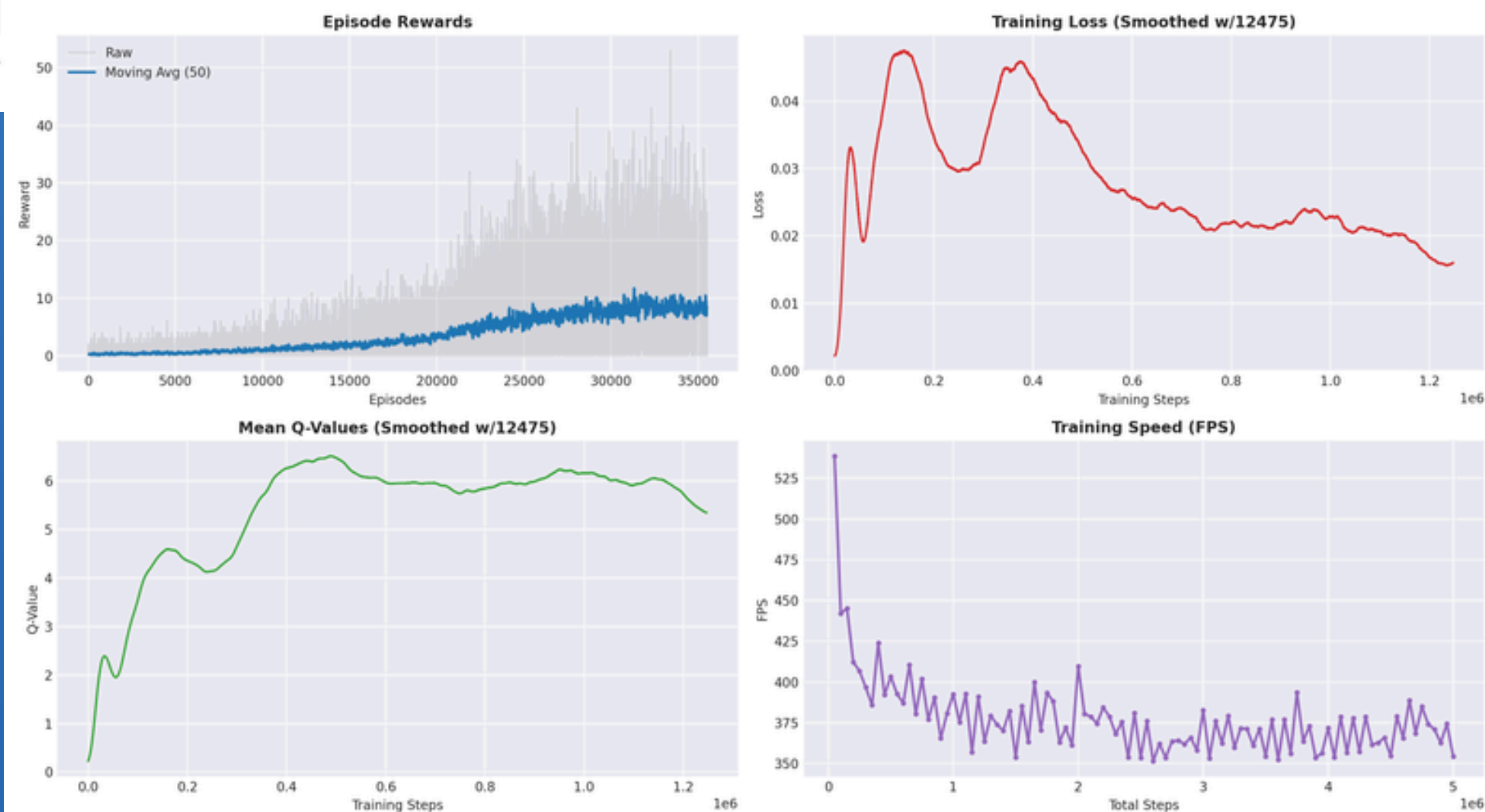


DQN Training Metrics: BreakoutNoFrameskip-v4



**lr=5e-5, bs=128,
eps_decay=1_000_000,
eps_end=0.1, buffer =200_000**

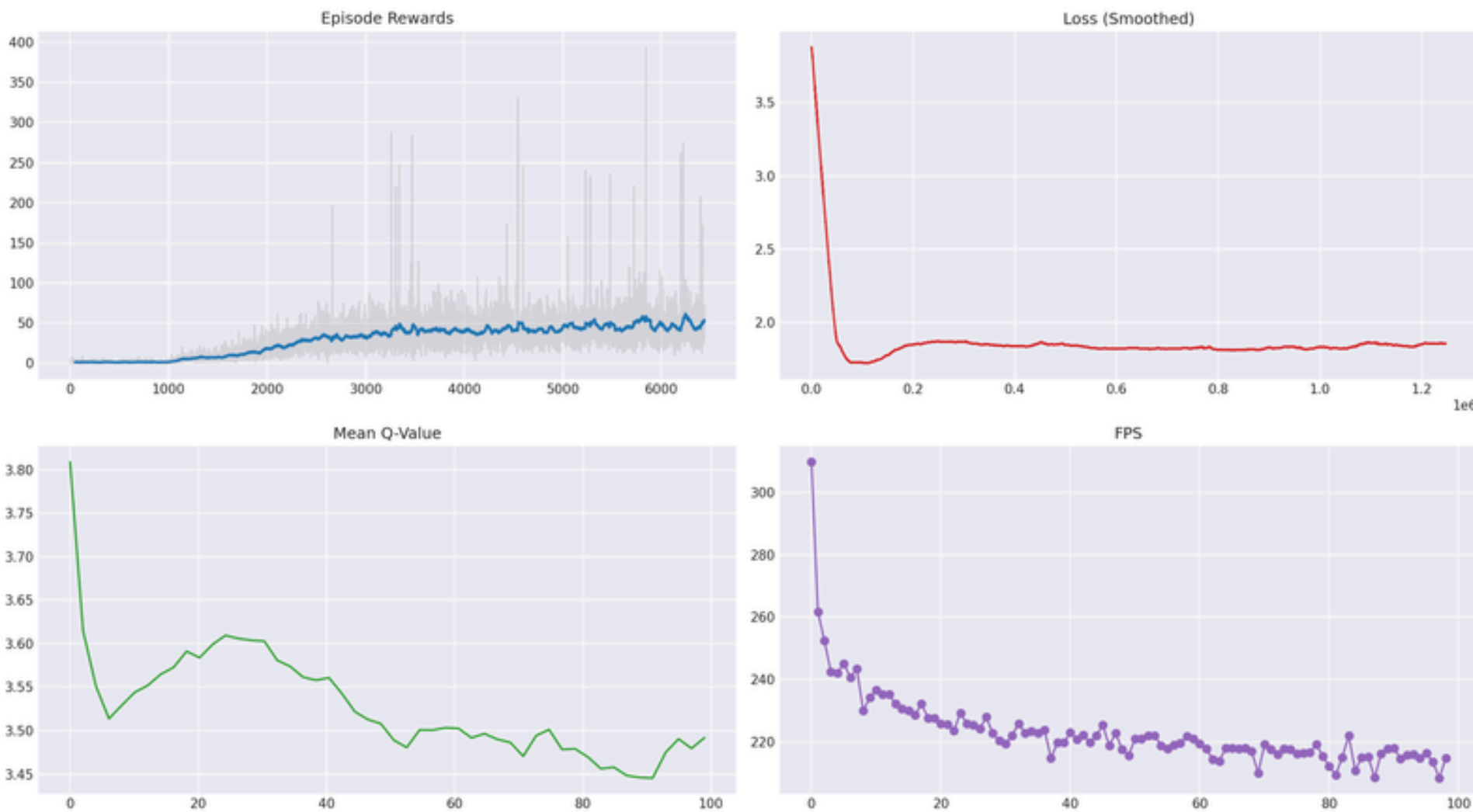
DQN Training Metrics: BreakoutNoFrameskip-v4



**lr = 5e-5, bs= 64, eps_decay =
1_000_000, eps_end=0.01,
buffer=200_000)**

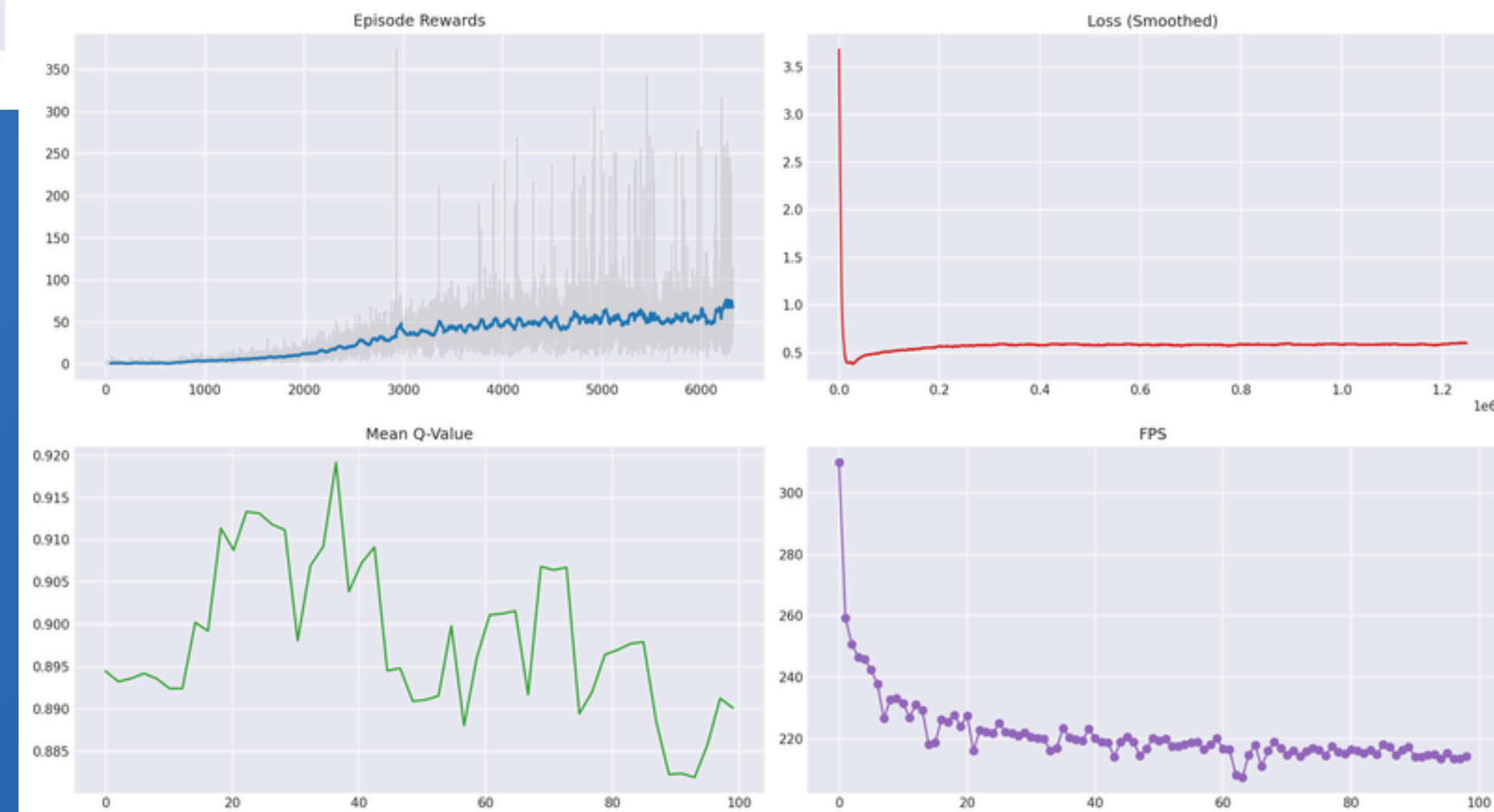
Breakout - DQN

C51 Metrics - 1_Baseline



Gamma = 0.90

C51 Metrics - 3_Gamma_090

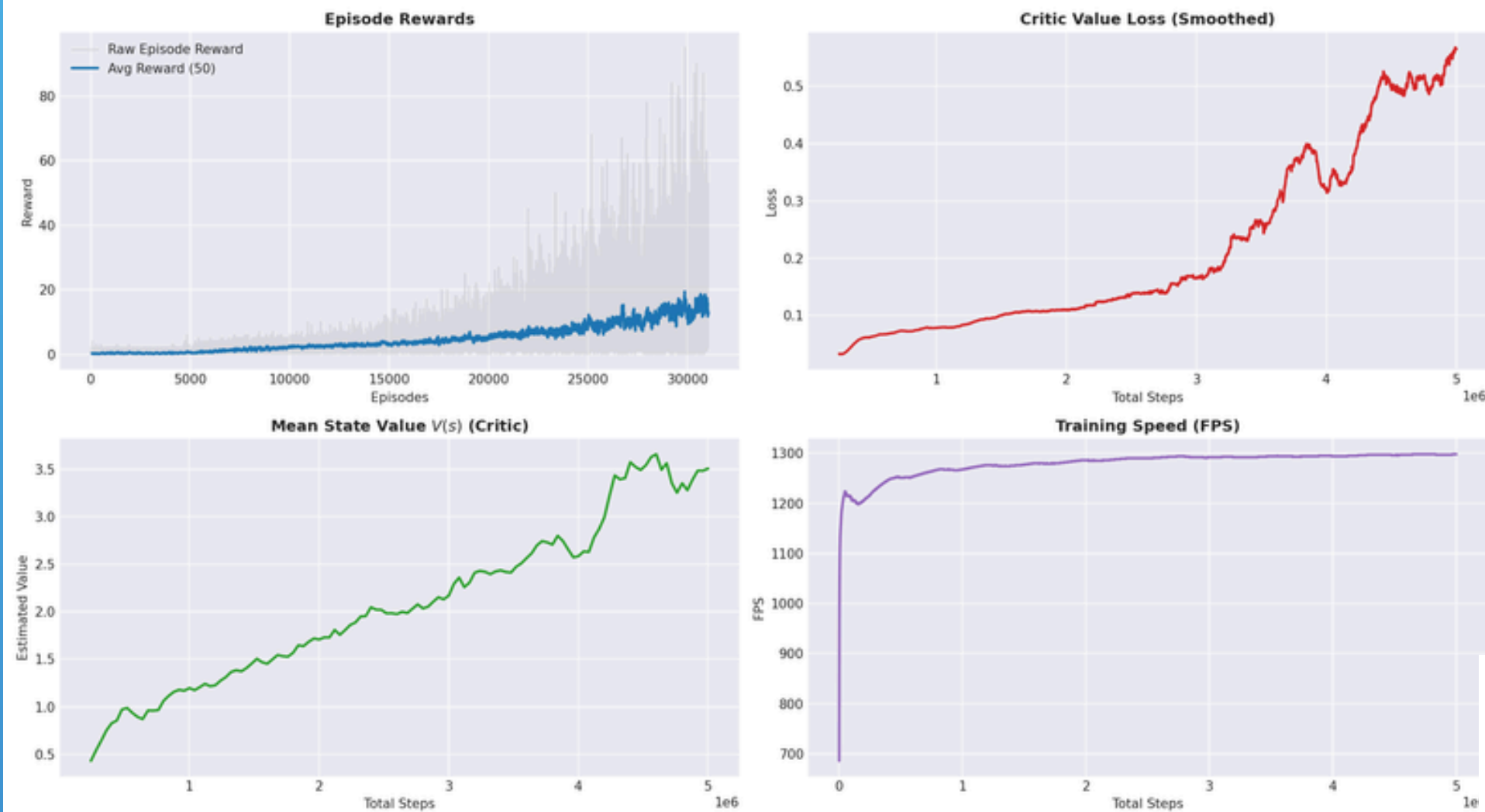


Baseline

"n_atoms": 51, "gamma": 0.99, "batch_size": 64,
 "lr": 2.5e-4 "buffer_size": 100_000,
 "min_replay": 10_000, "target_tau": 0.005

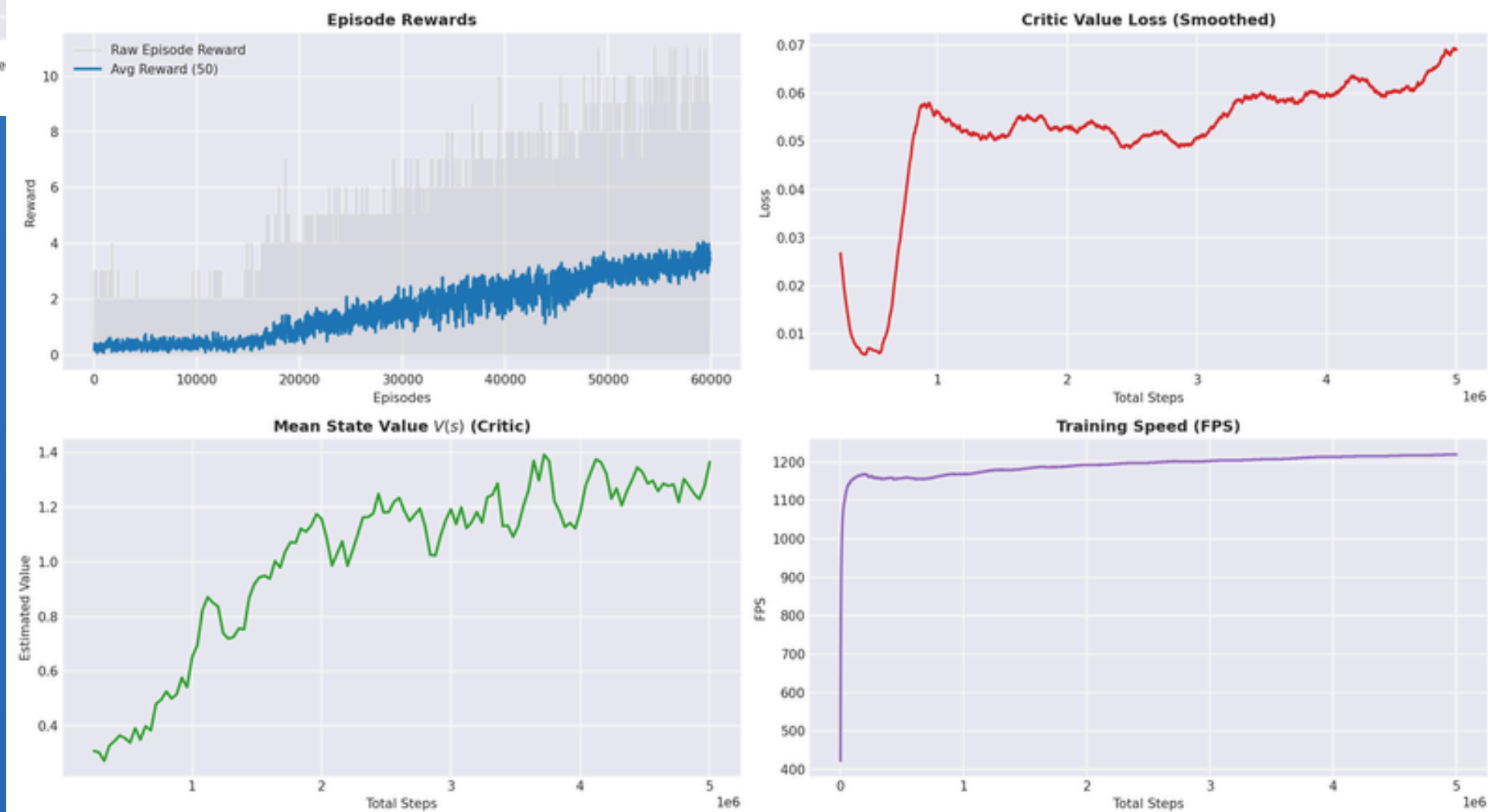
Breakout - C51

PPO Training Metrics: BreakoutNoFrameskip-v4



`"lr": 5e-5, "batch_size": 256, "n_steps": 128, "ent_coef": 0.01`

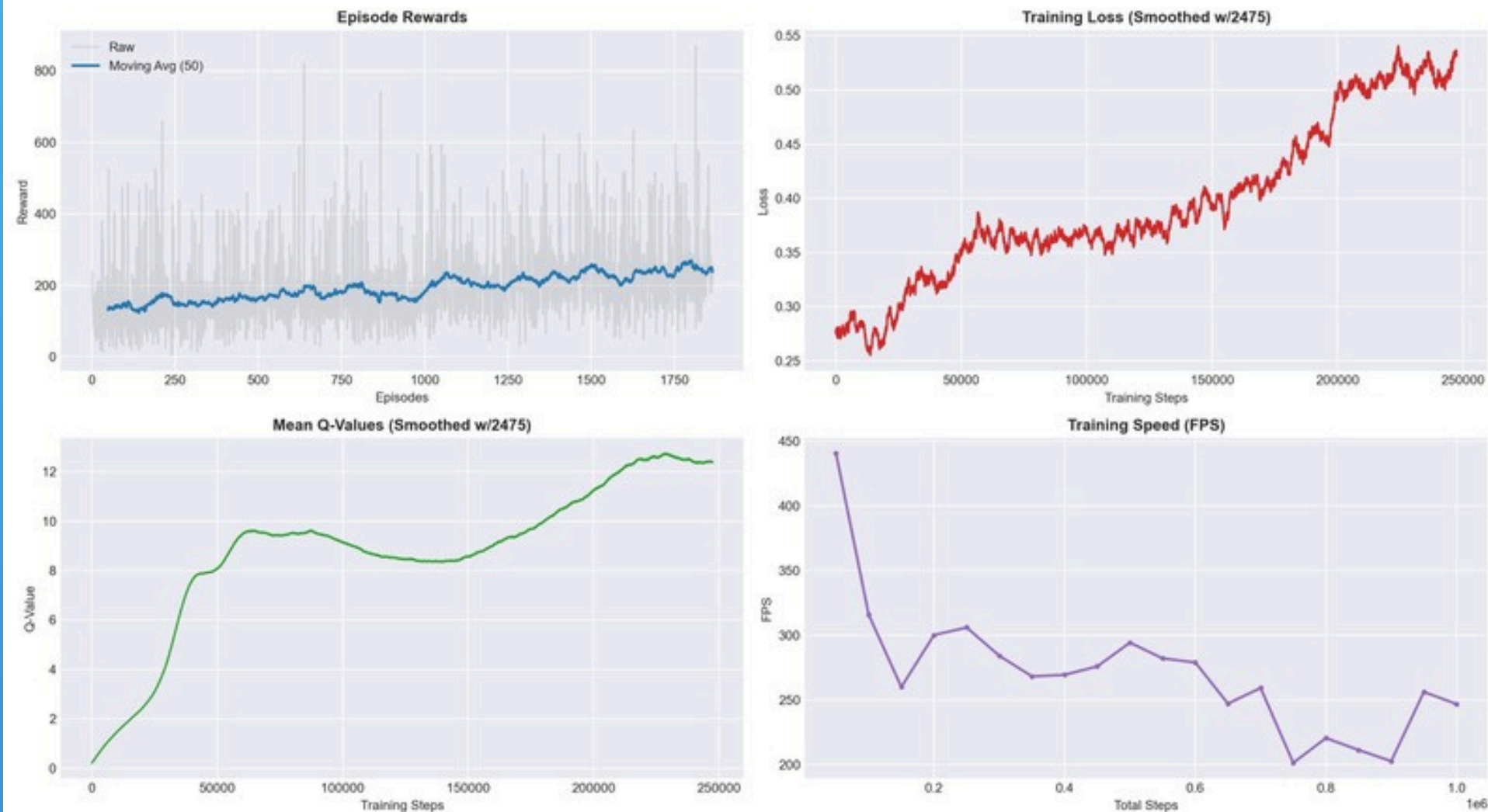
PPO Training Metrics: BreakoutNoFrameskip-v4



`"lr": 2.5e-4, "batch_size": 512, "n_steps": 128, "ent_coef": 0.01`

Breakout - PPO

DQN Training Metrics: SpaceInvadersNoFrameskip-v4

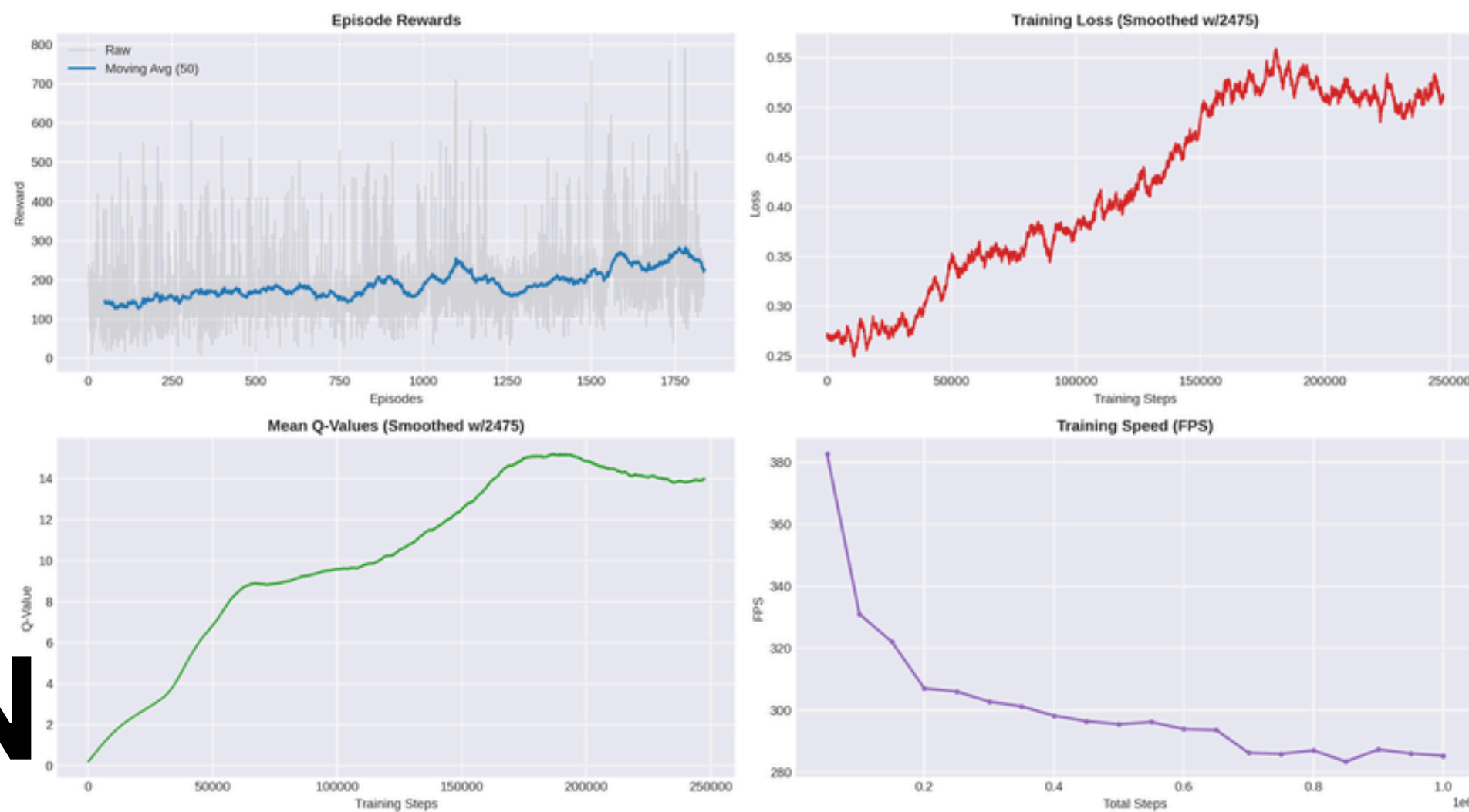


**lr=5e-5, bs=64,
eps_decay_frames=1_000_000,
eps_end=0.1,
buffer_size=300_000**

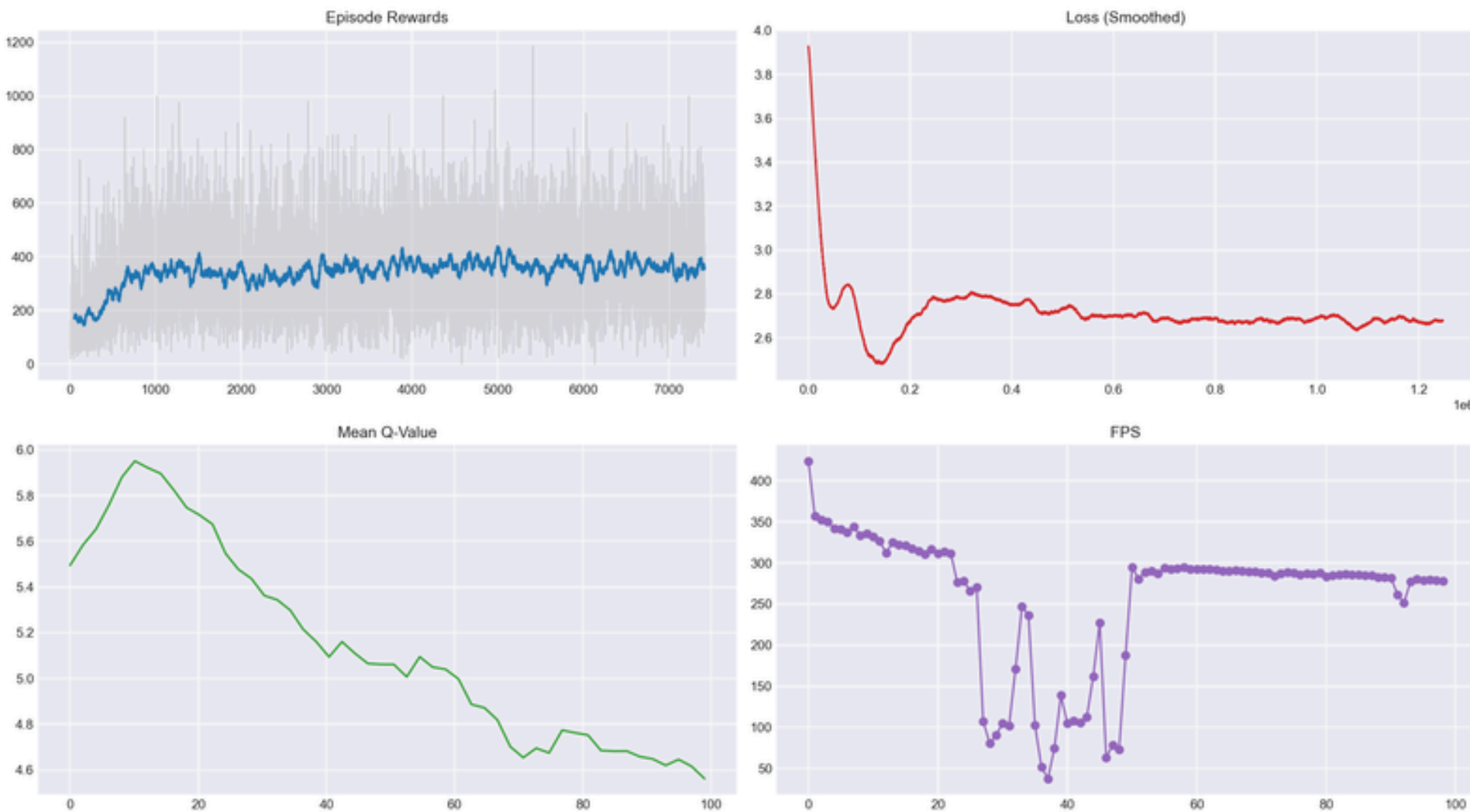
**lr=2.5e-4, bs=64, 1_000_000,
0.1, buffer_size=200_000),**

Space Invaders - DQN

DQN Training Metrics: SpaceInvadersNoFrameskip-v4

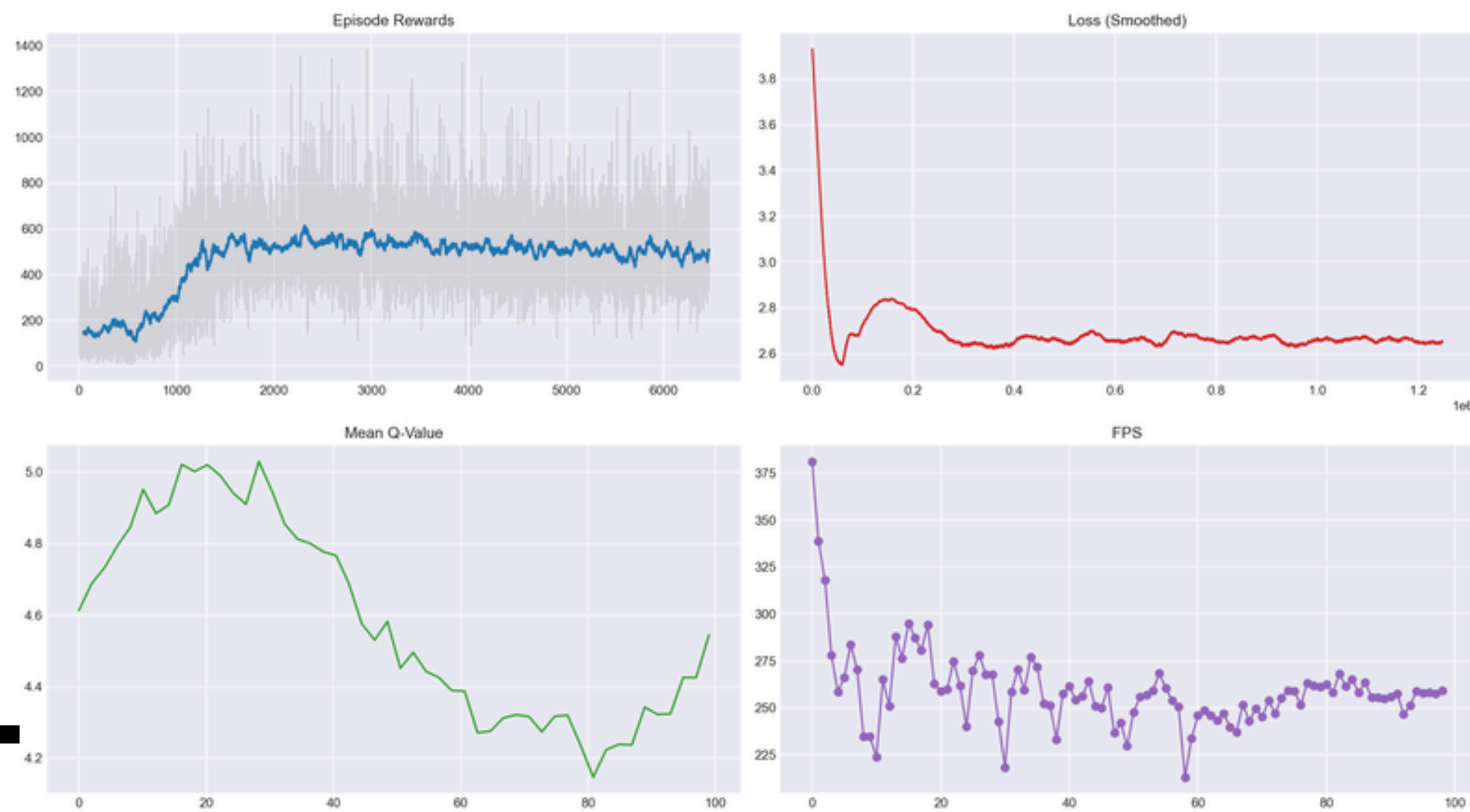


C51 Metrics - 1_Baseline



5. LR = 1e-4

C51 Metrics - 5_LR_1e-4

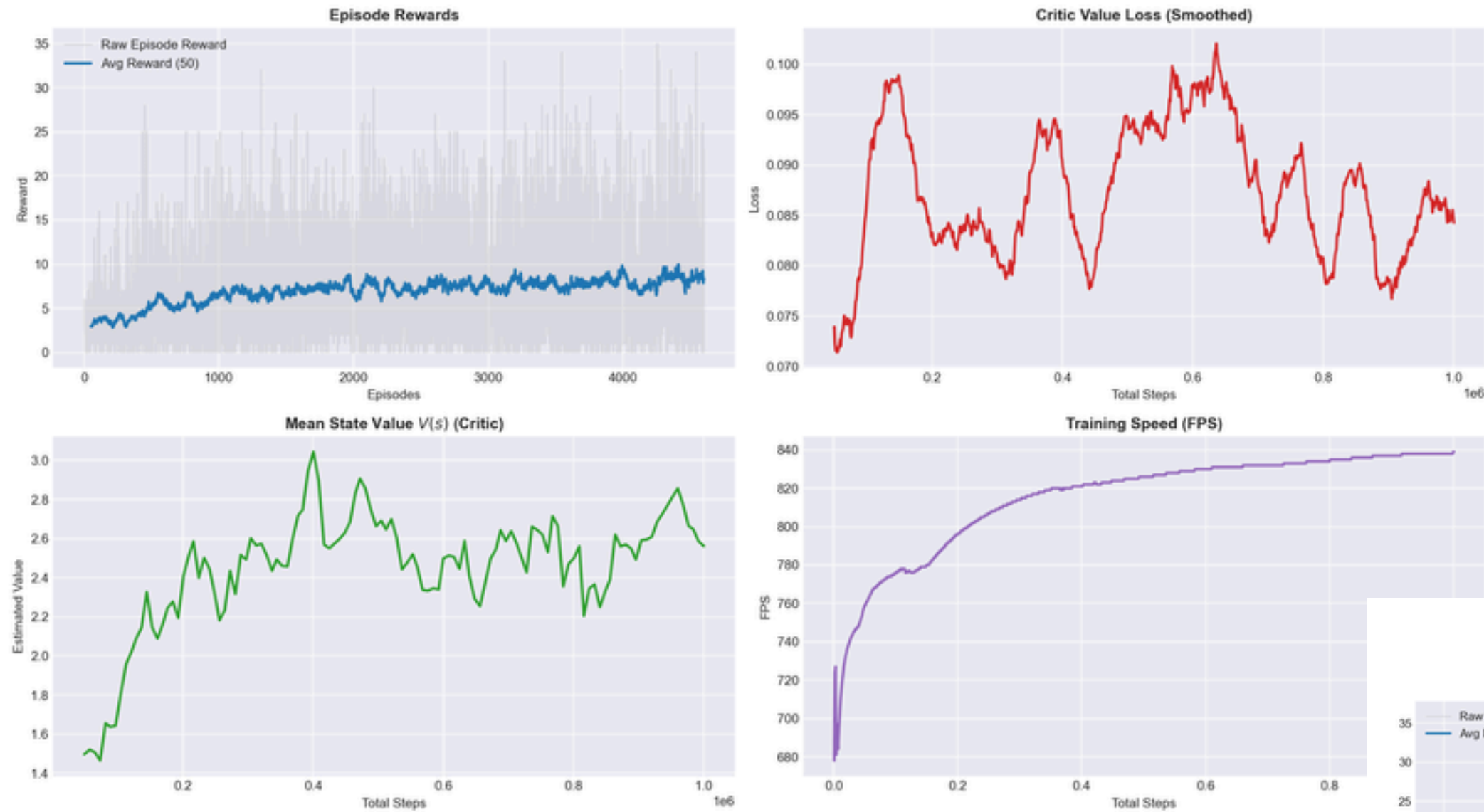


Baseline

"n_atoms": 51, "gamma": 0.99, "batch_size": 64, "lr": 2.5e-4 "buffer_size": 100_000, "min_replay": 10_000, "target_tau": 0.005

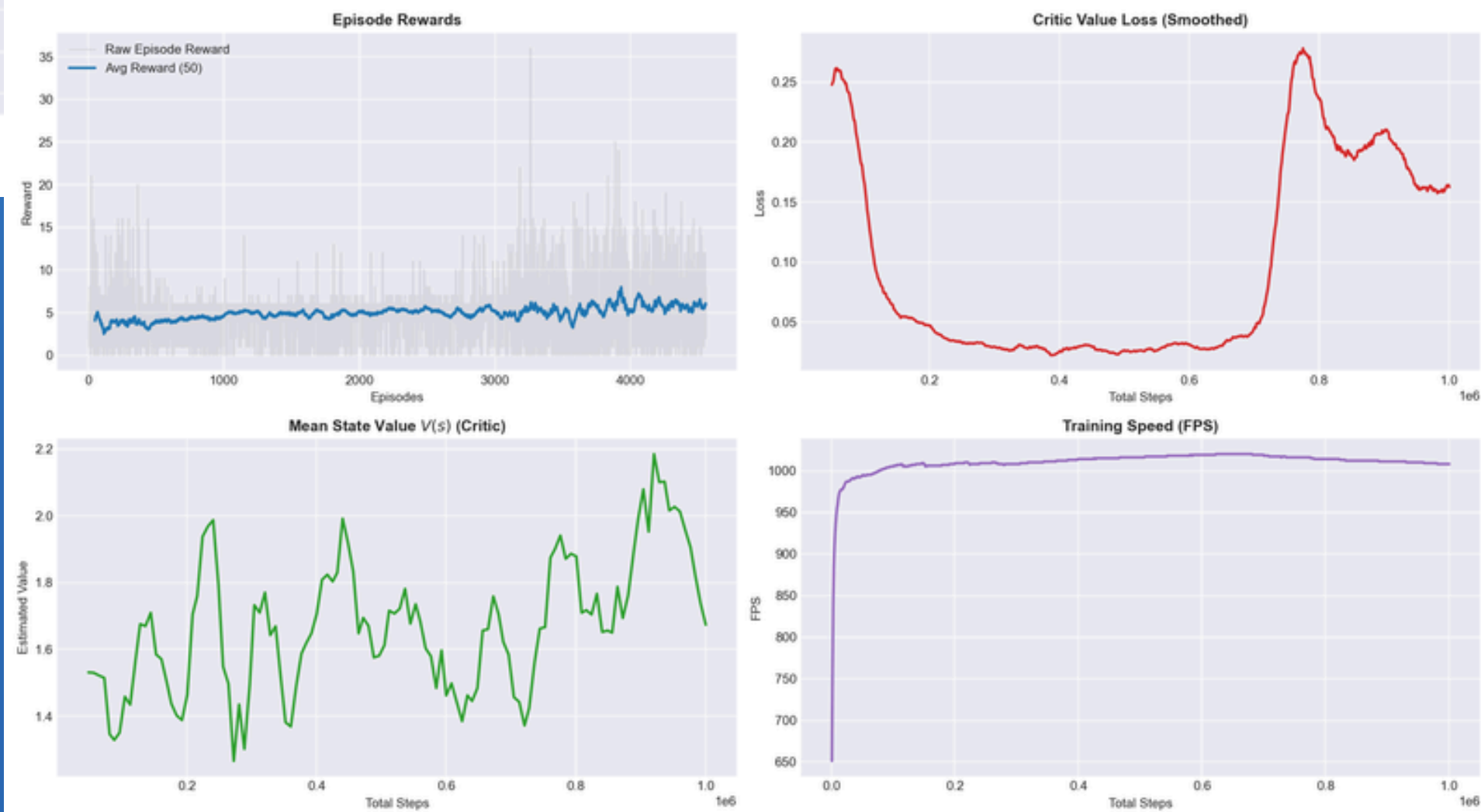
Space Invaders-C51

PPO Training Metrics: SpaceInvadersNoFrameskip-v4



large batch

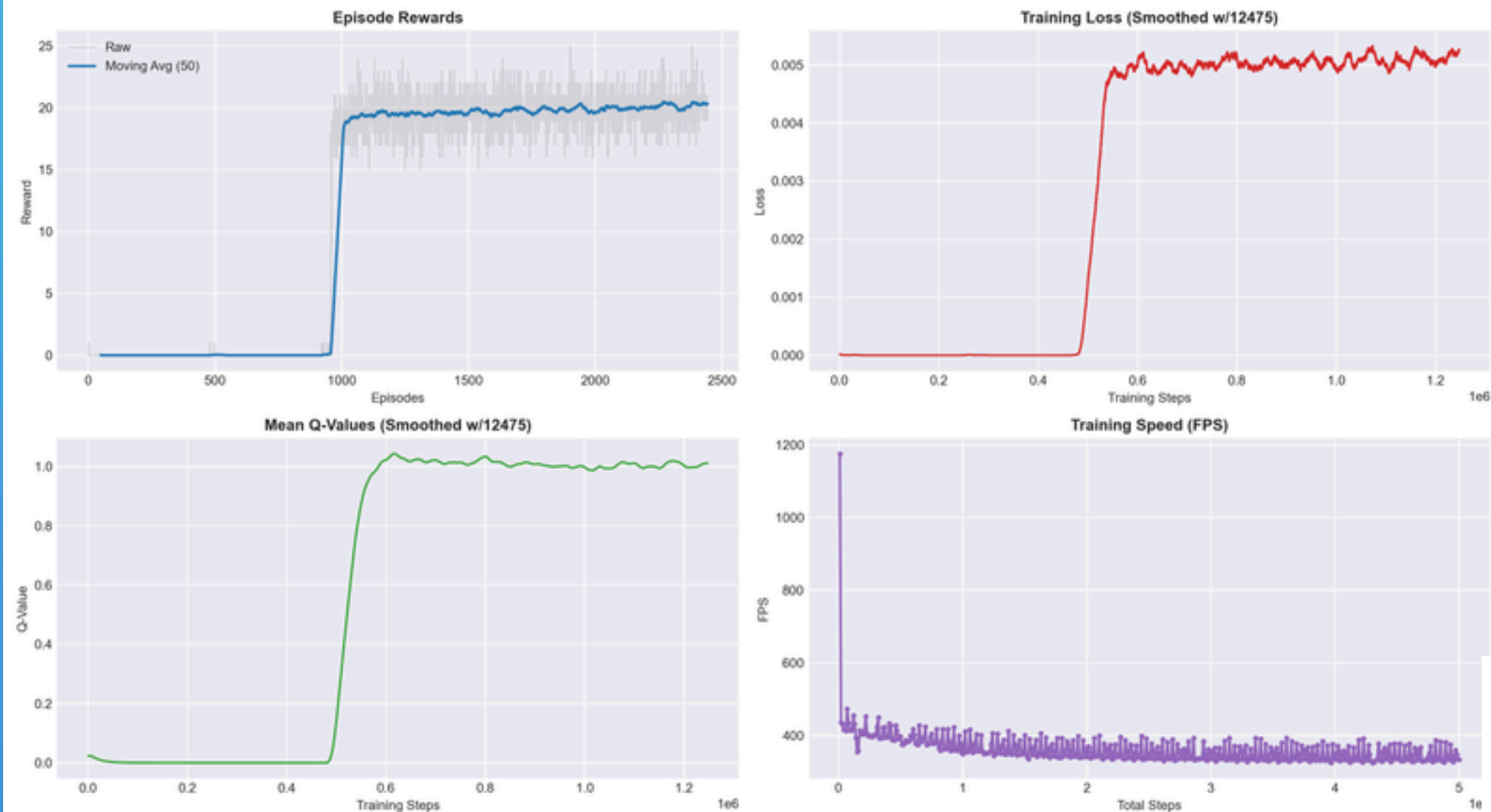
PPO Training Metrics: SpaceInvadersNoFrameskip-v4



small batch

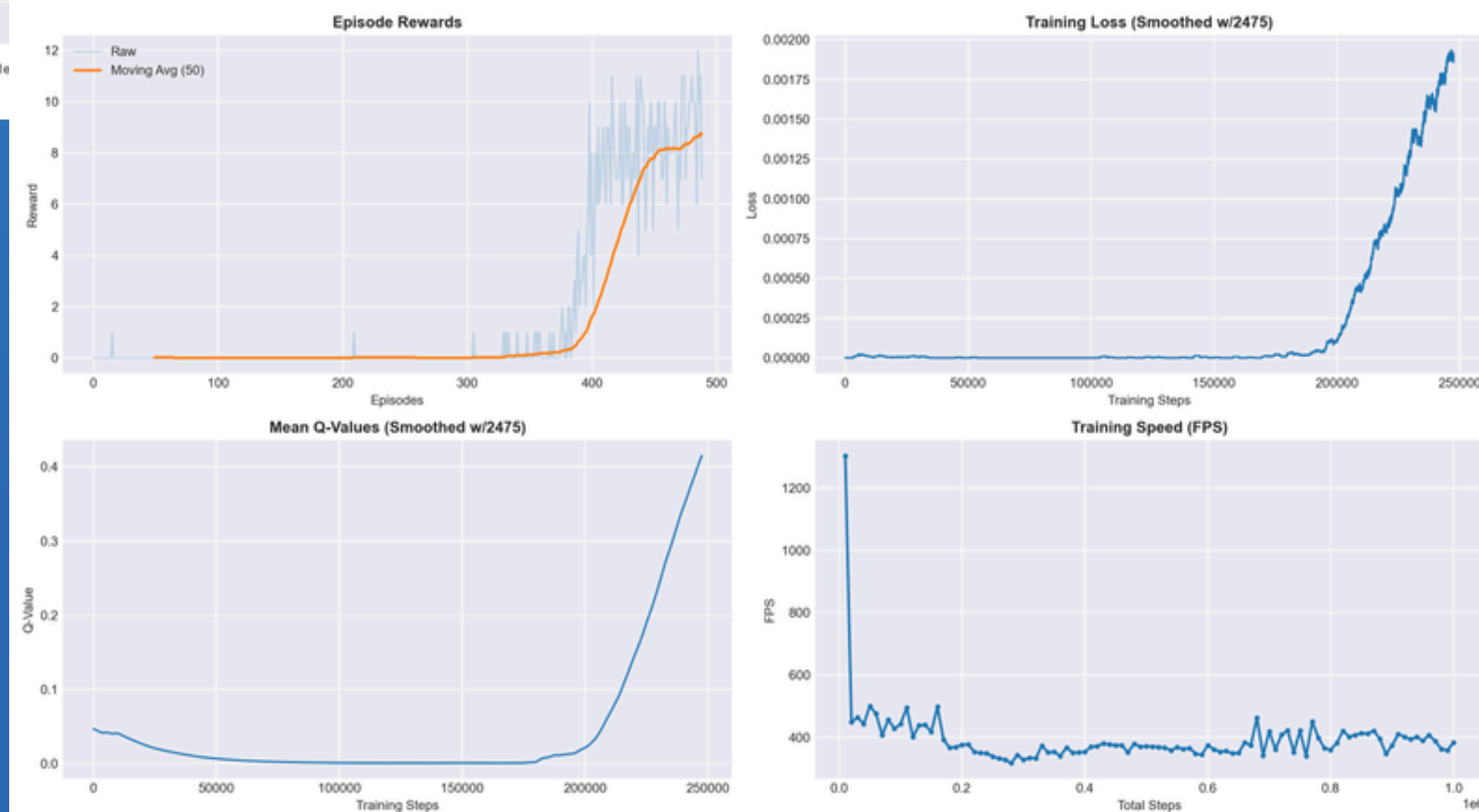
Space Invaders
- PPO

DQN Training Metrics: FreewayNoFrameskip-v4 (Duration: 3:51:52)



2. 5e-5, 64, 1_500_000,
0.1, 200_000

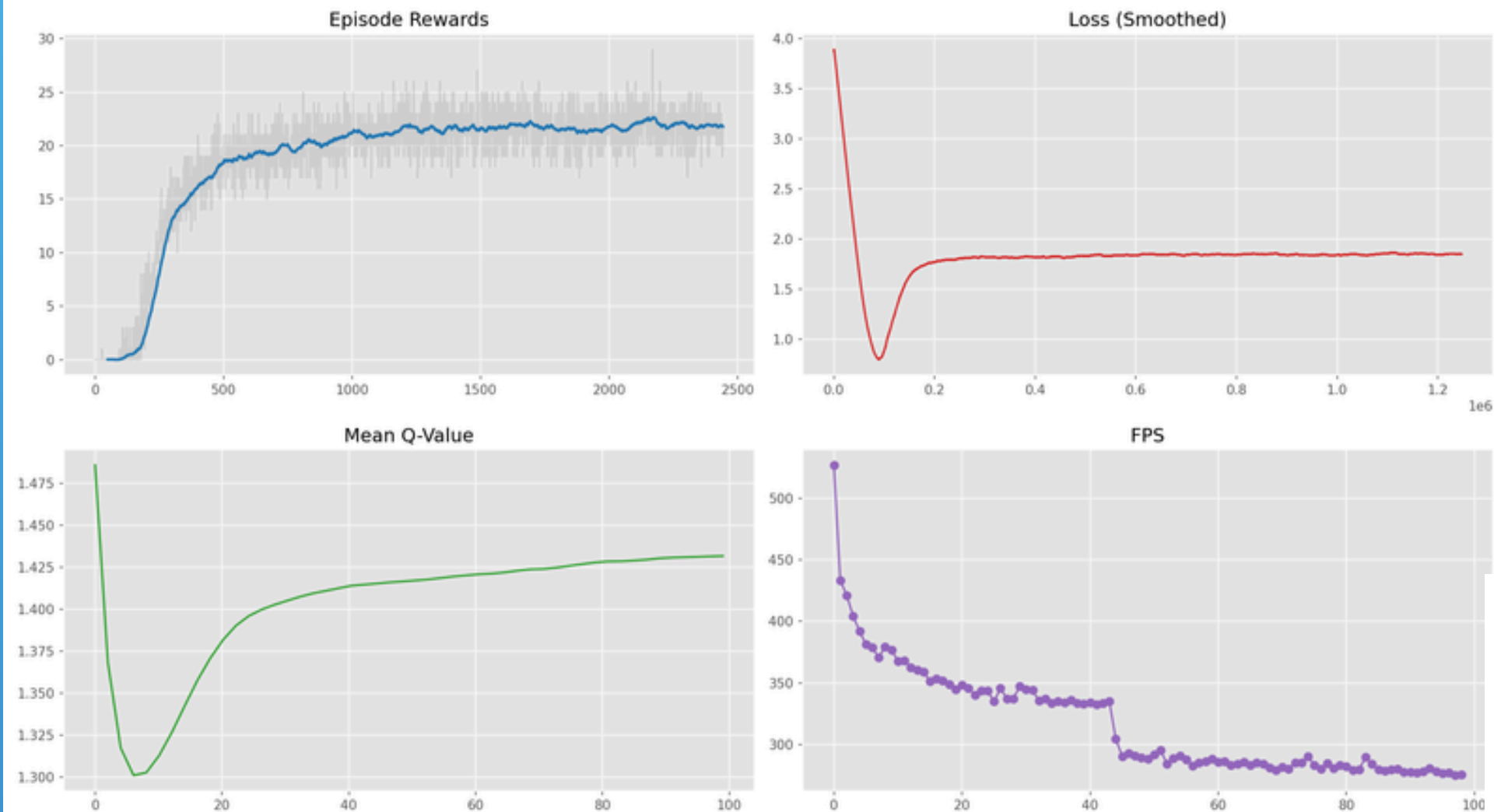
DQN Training Metrics: FreewayNoFrameskip-v4 (Duration: 0:43:27)



1. 1e-4,32,1_000_000,0.1,
200_000

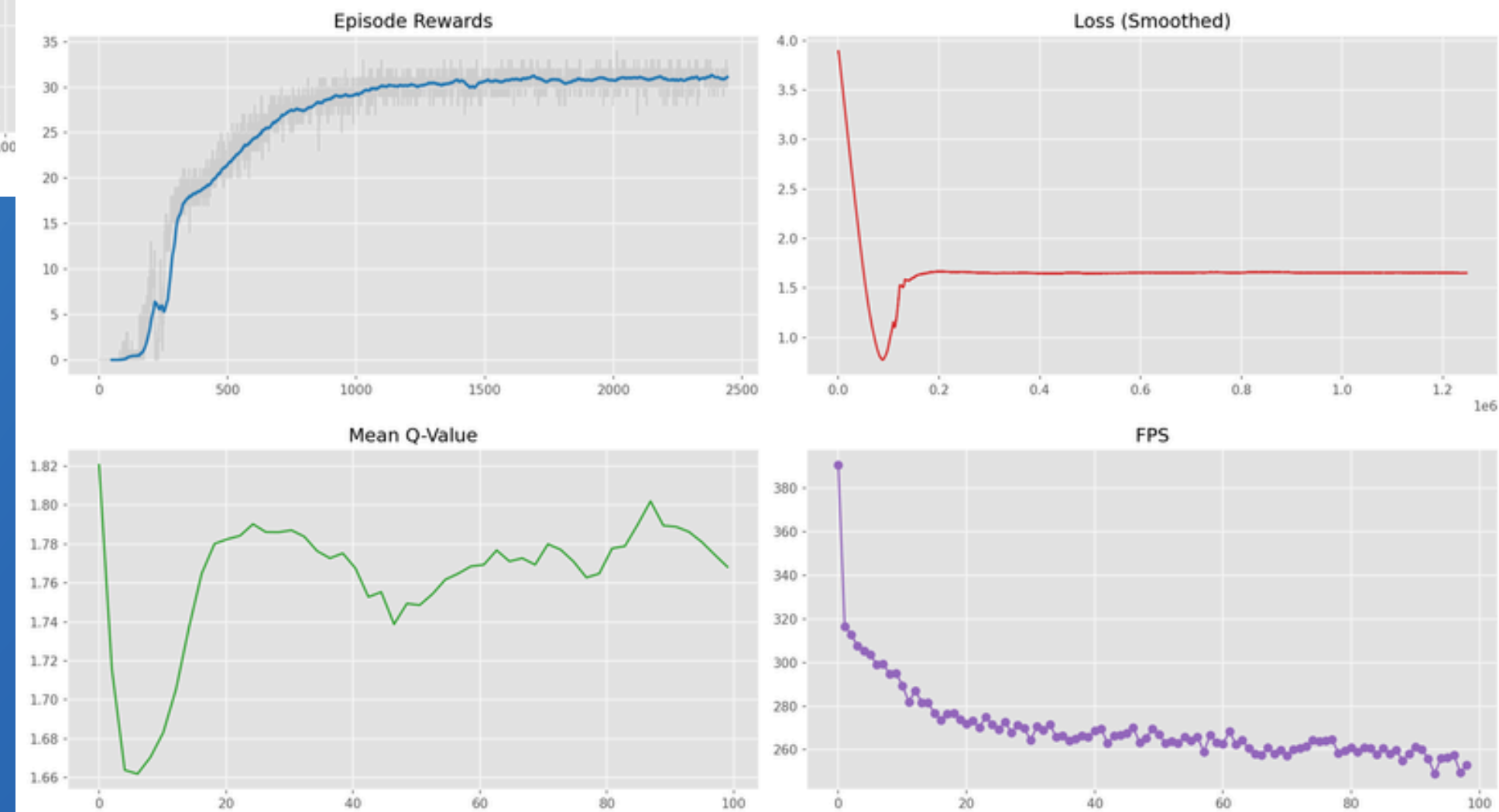
Freeway - DQN

C51 Metrics - 1_Baseline



4. Batch Size = 128

C51 Metrics - 4_Batch_128

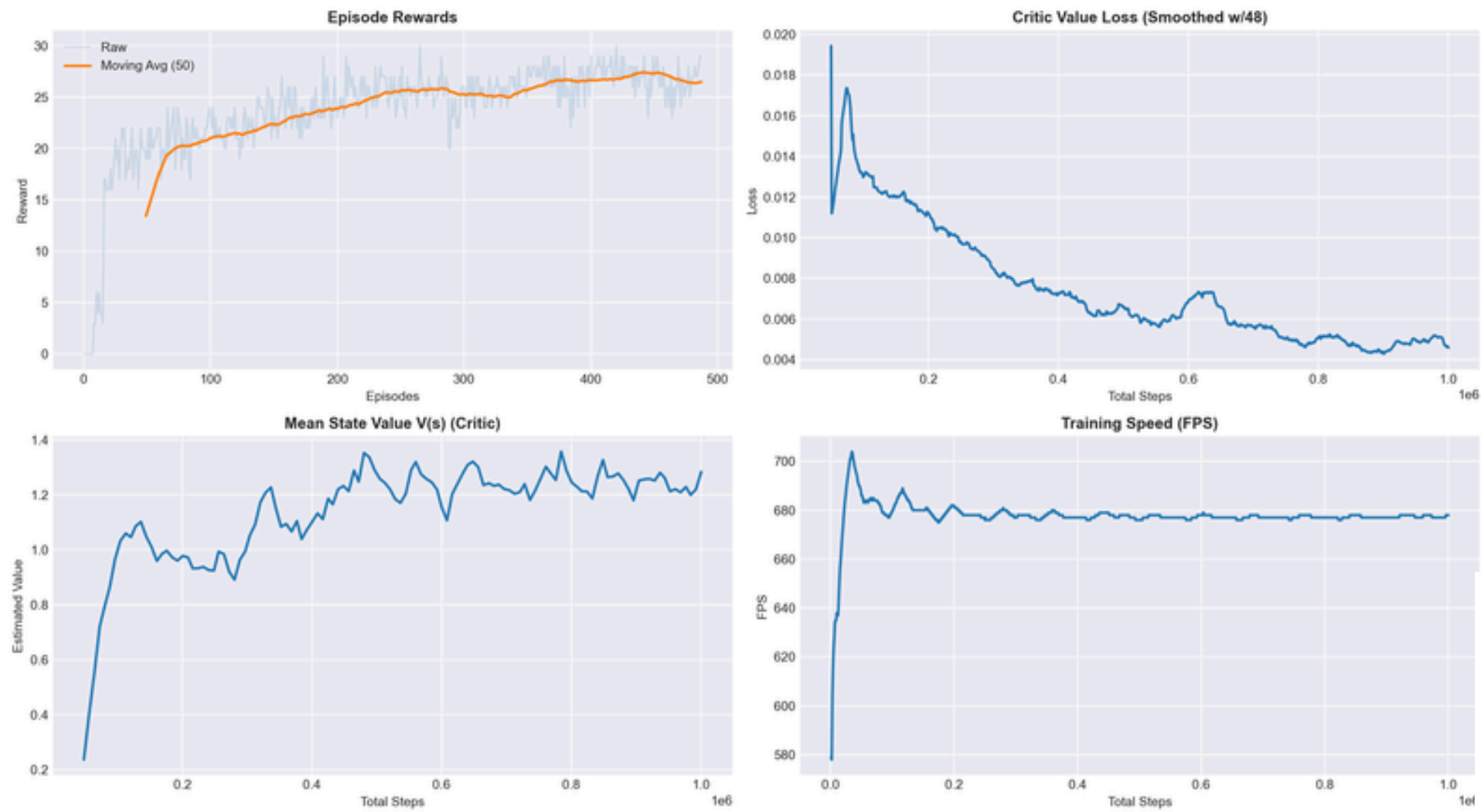


Baseline

"n_atoms": 51, "gamma": 0.99, "batch_size":
64, "lr": 2.5e-4 "buffer_size": 100_000,
"min_replay": 10_000, "target_tau": 0.005

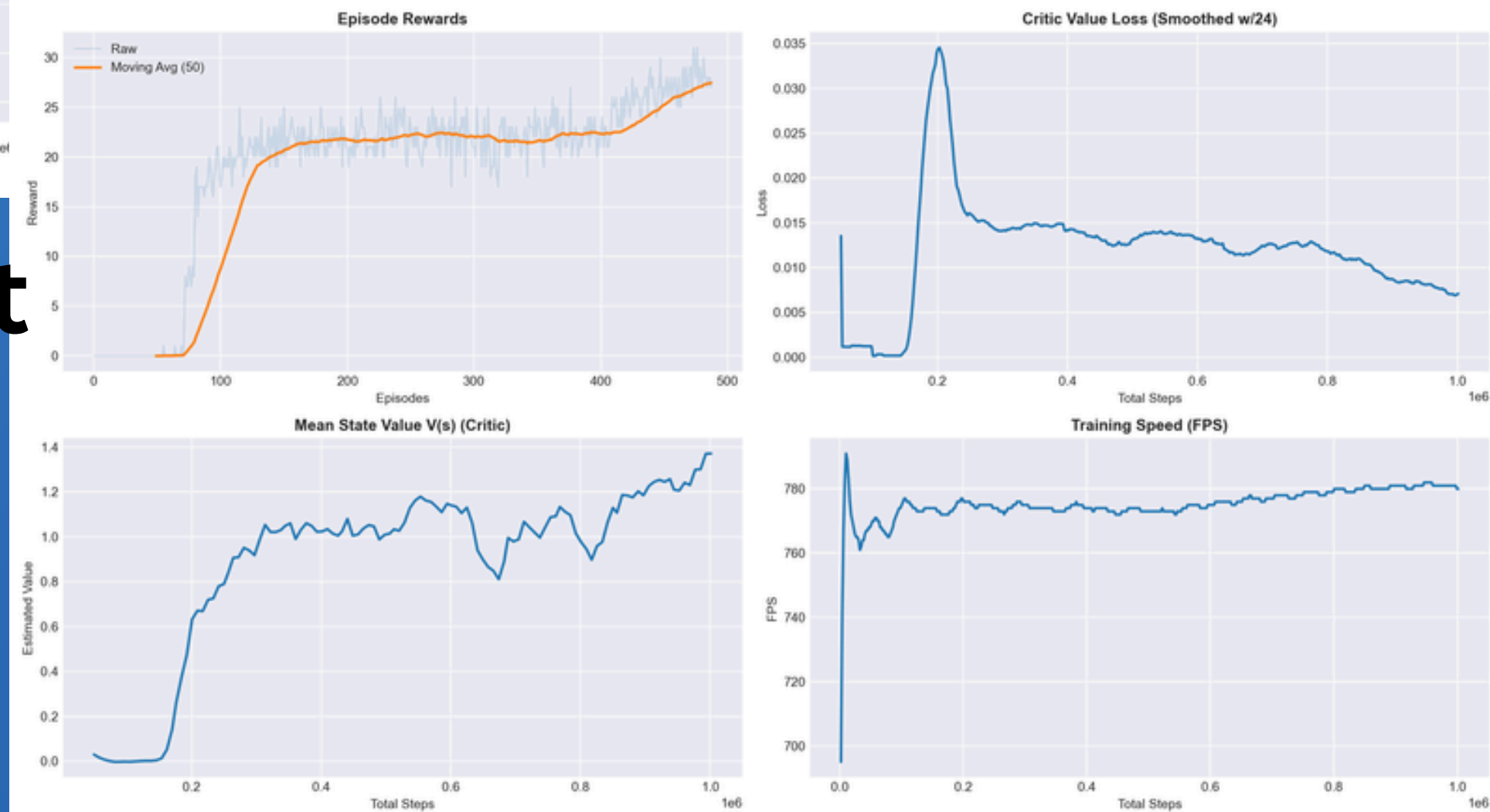
Freeway - C51

PPO Training Metrics: FreewayNoFrameskip-v4 | lr=0.00025, batch_size=64, n_steps=128, ent_coef=0.01 | Run run_5



5. Lr=0.00025_batch=256_
nsteps=256_ent=0.01

PPO Training Metrics: FreewayNoFrameskip-v4 | lr=0.00025, batch_size=256, n_steps=256, ent_coef=0.01 | Run run_4

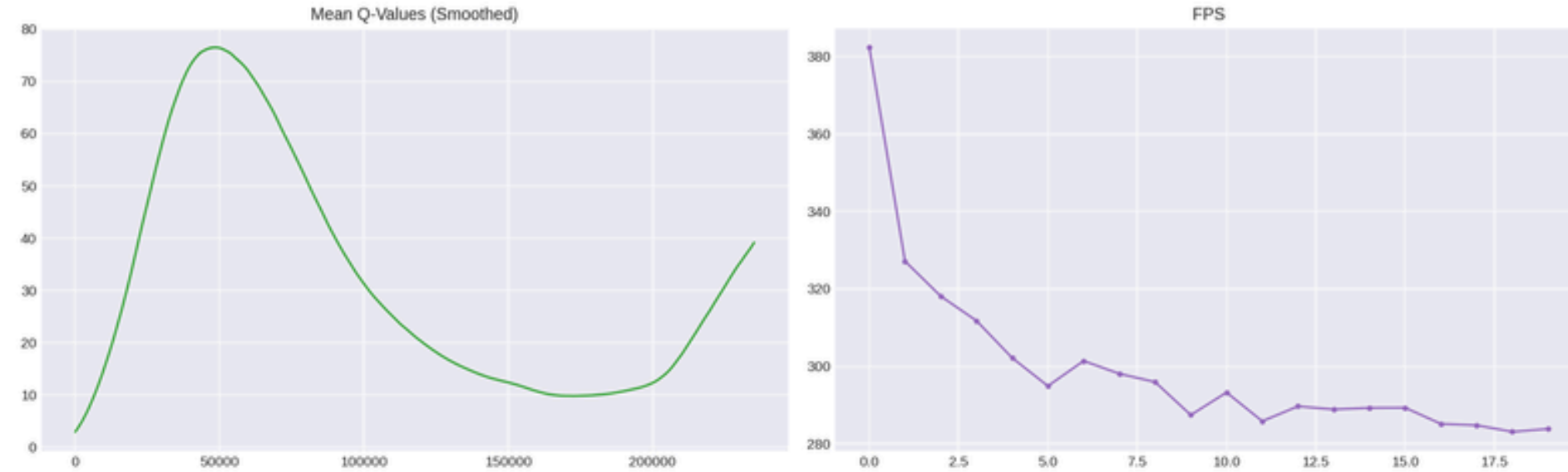
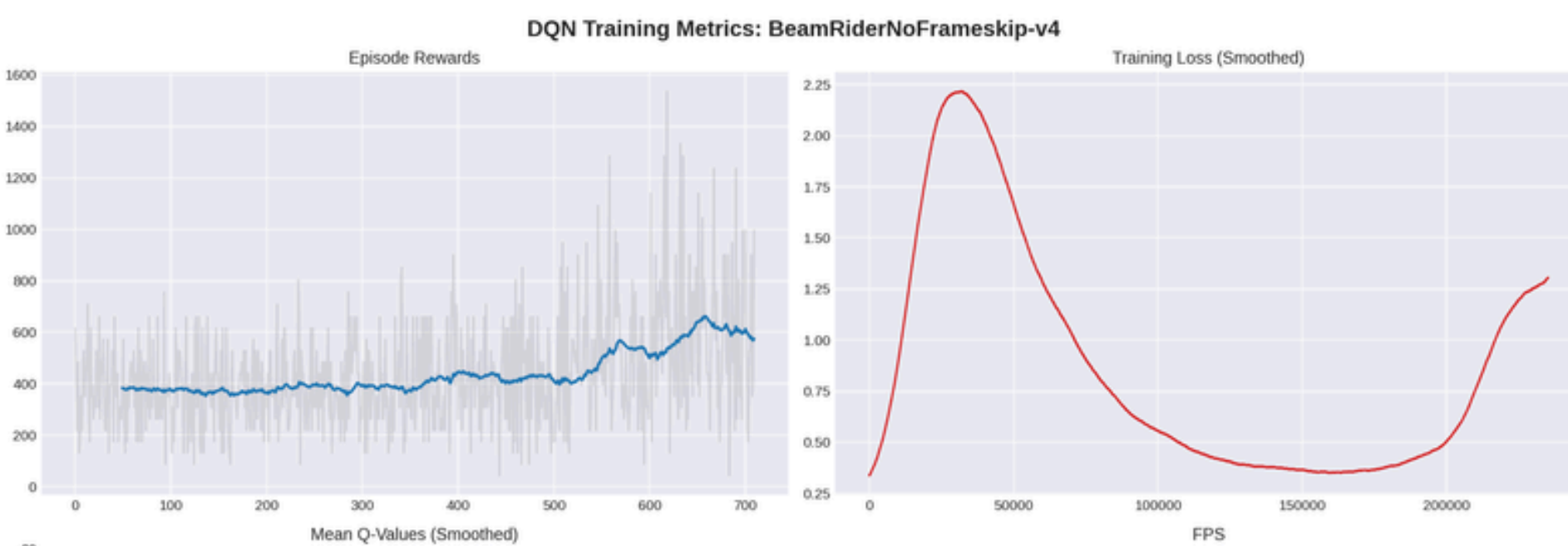


1. Lr=0.00025_batch=64_nst
eps=128_ent=0.01

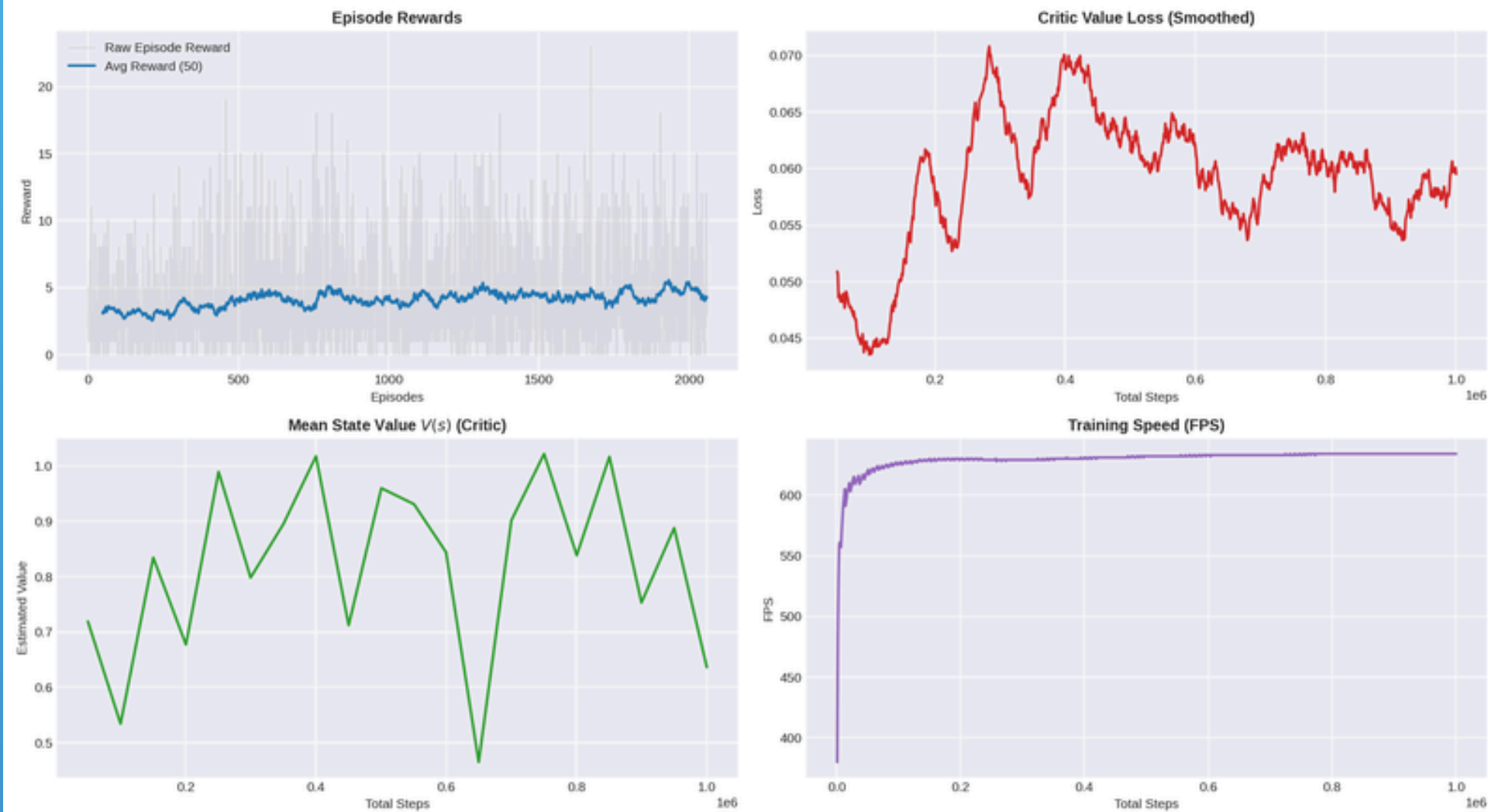
Freeway - PPO

5. LR5e-05_BS64_Decay1000000_Buf3000000

1. LR0.00025_BS64_Decay1000000_Buf2000000
Beam Rider
- DQN



PPO Training Metrics: BeamRiderNoFrameskip-v4

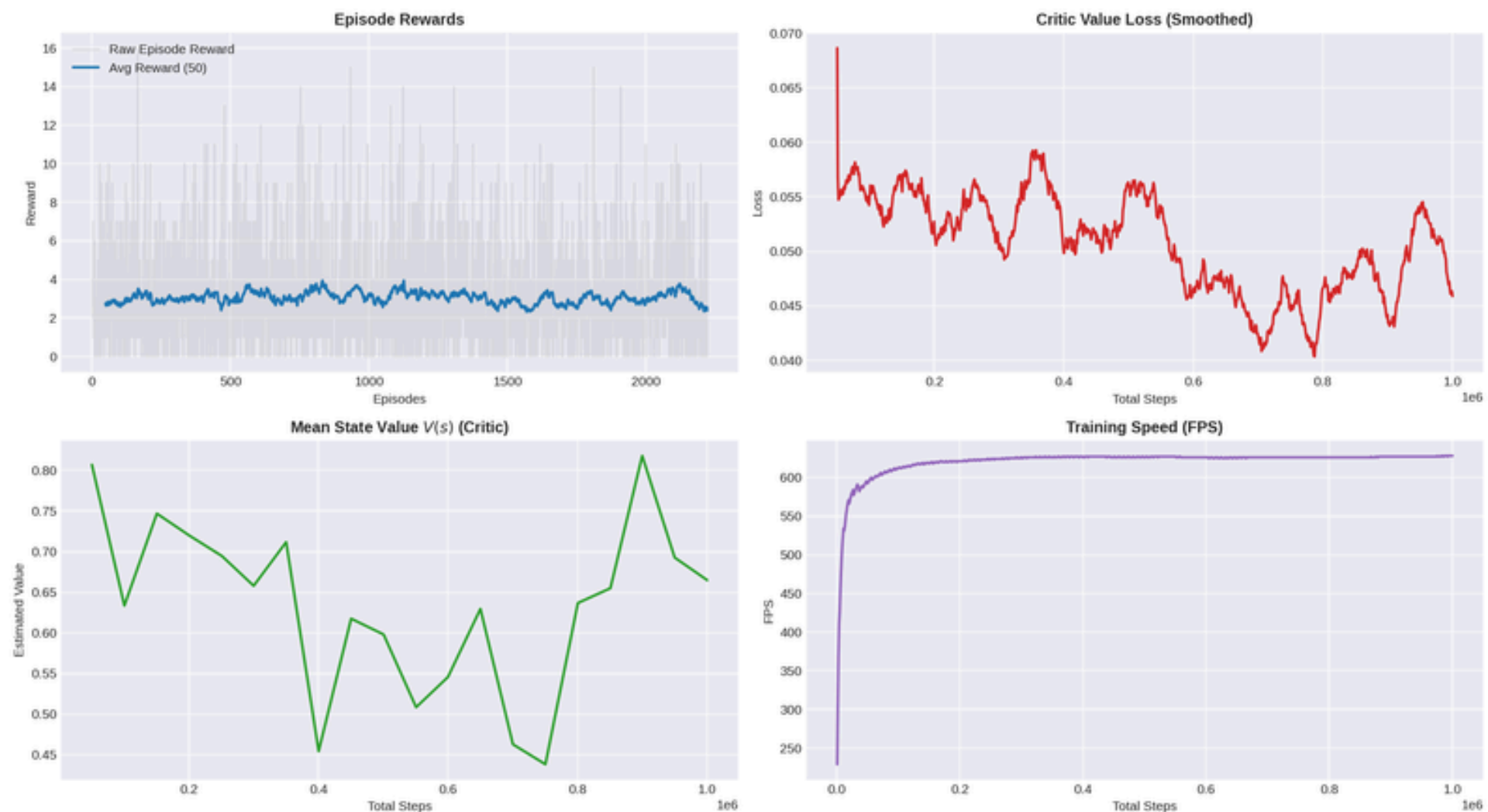


Lr=0.00025_batch=64
nsteps=128_ent=0.01

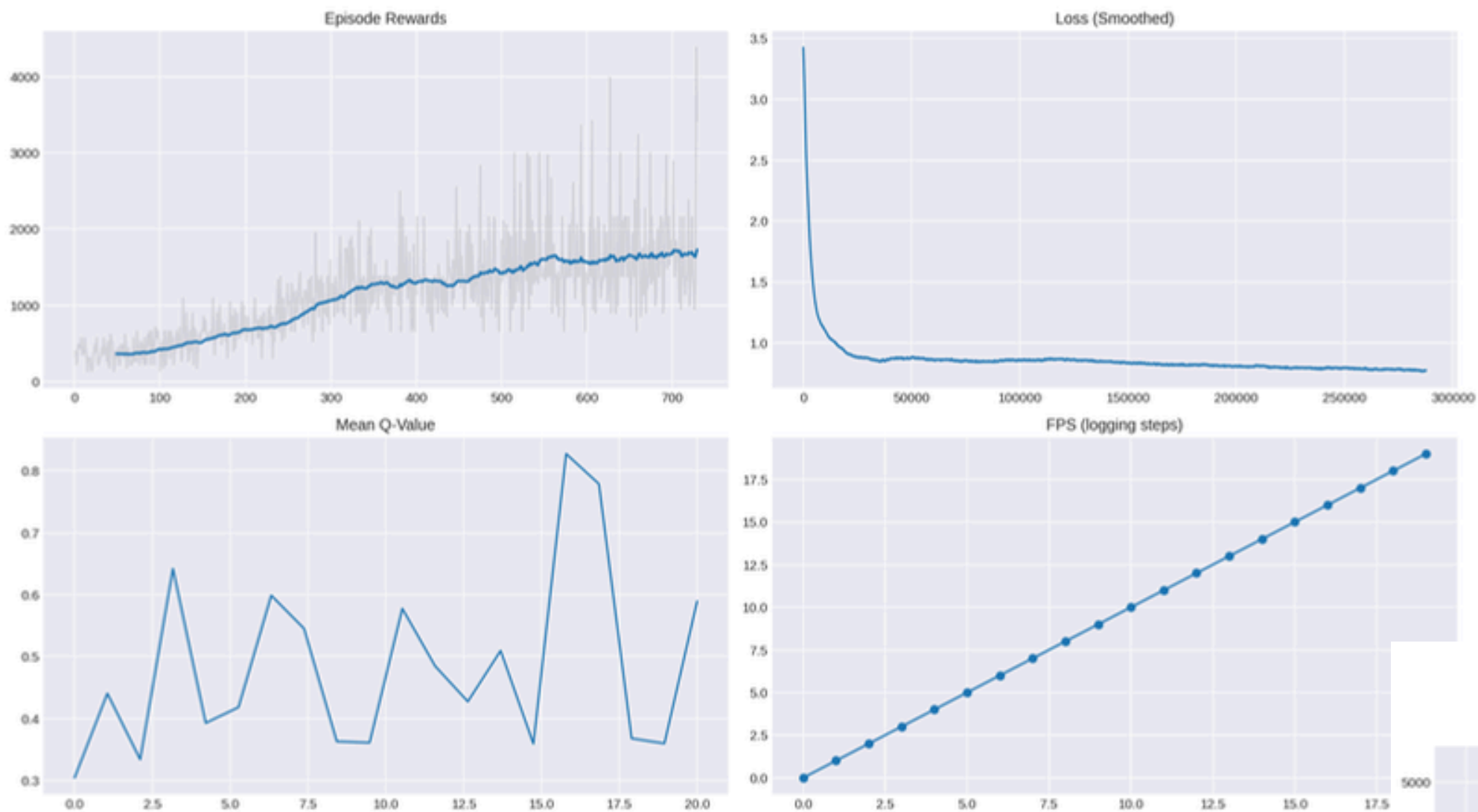
Beam Rider-PPO

3.lr0.001_batch256

PPO Training Metrics: BeamRiderNoFrameskip-v4

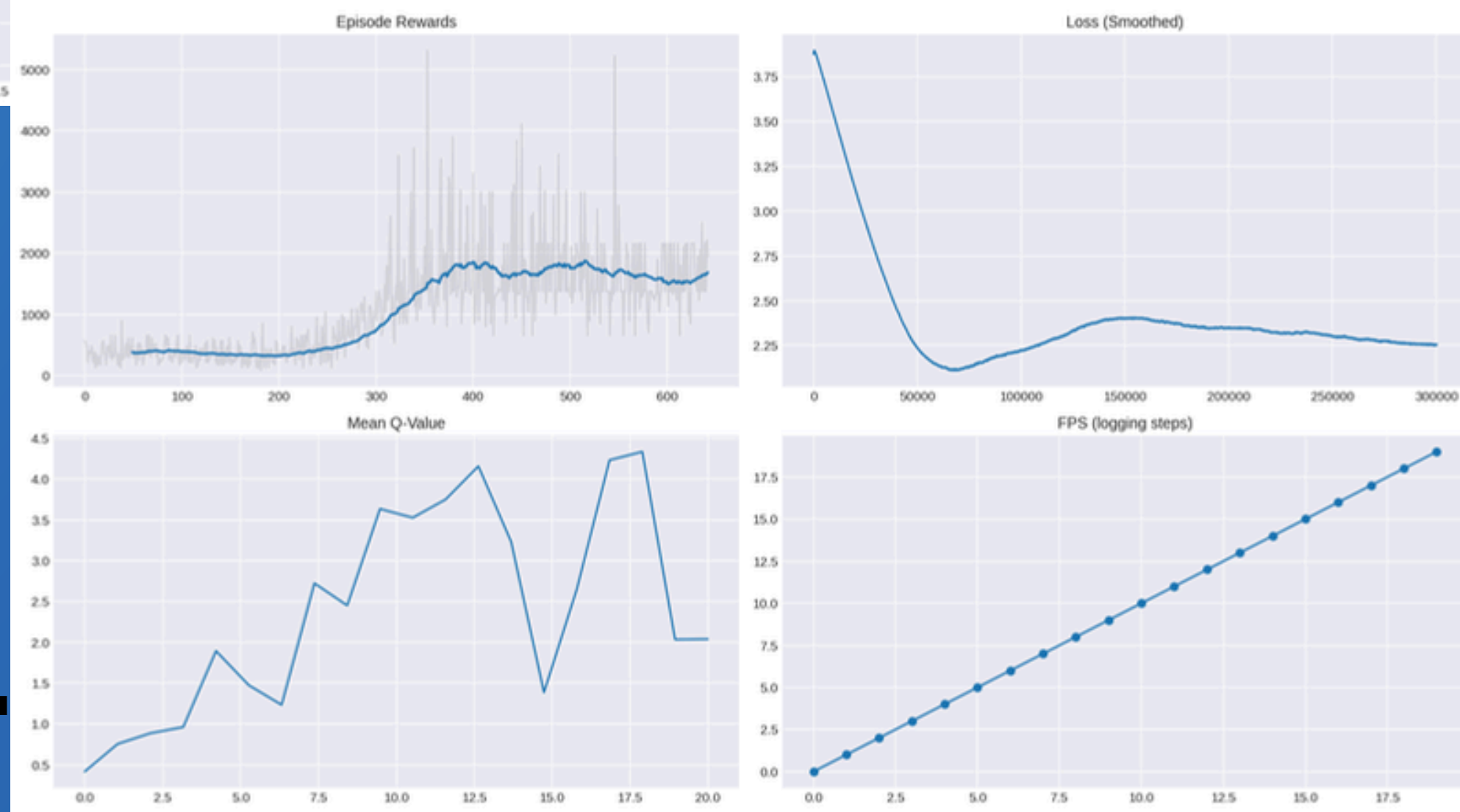


C51 Metrics – BeamRider (Gamma = 0.9)



5. LR = $1e-4$

C51 Metrics – BeamRider (LR = $1e-4$)



"n_atoms": 51,"gamma":

0.90,"batch_size": 64,

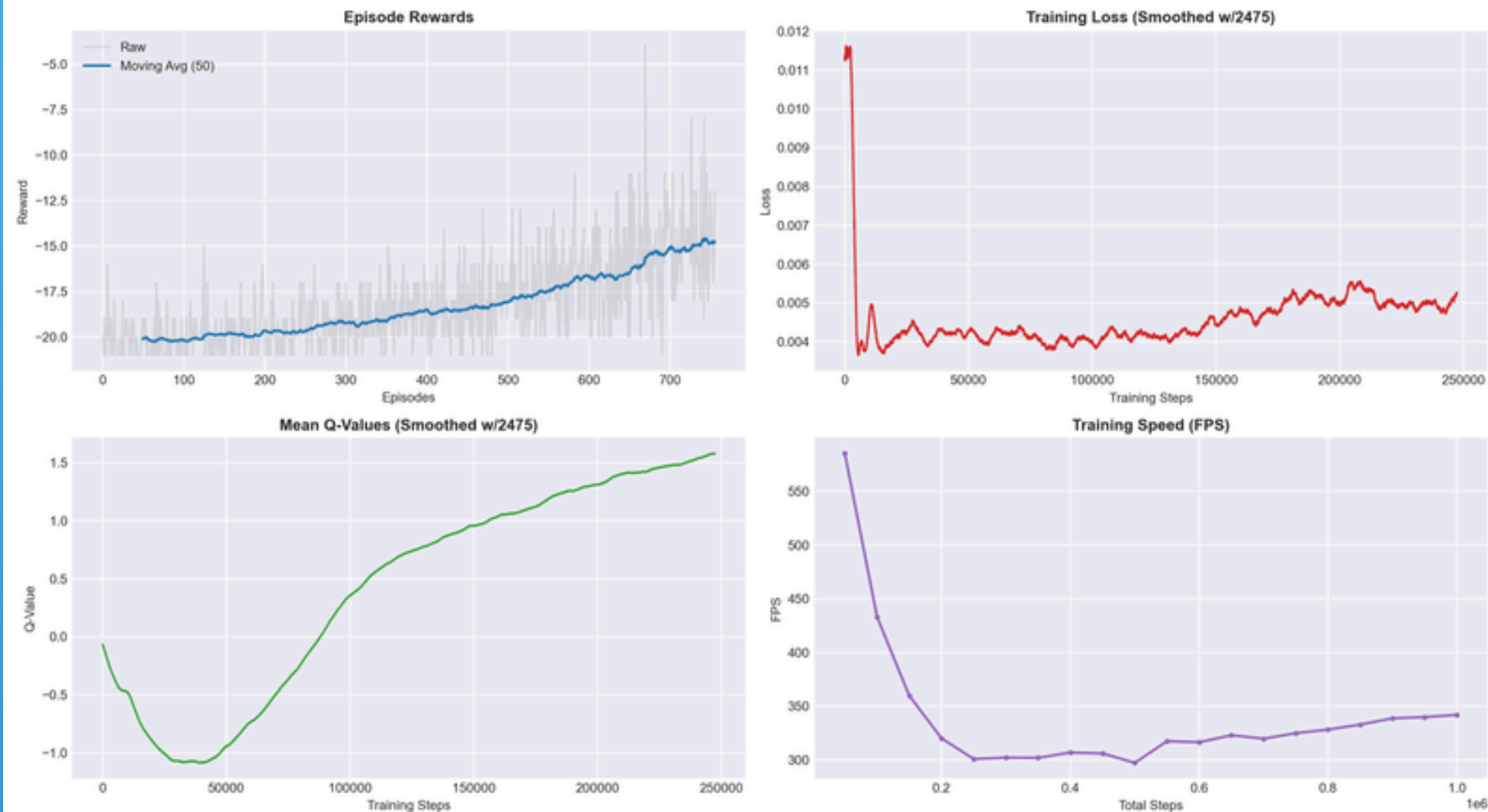
"lr": $2.5e-4$,"buffer_size":

100_000,"min_replay": 10_000,

"target_tau": 0.005, "train_every": 4

Beam Rider - C51

DQN Training Metrics: PongNoFrameskip-v4

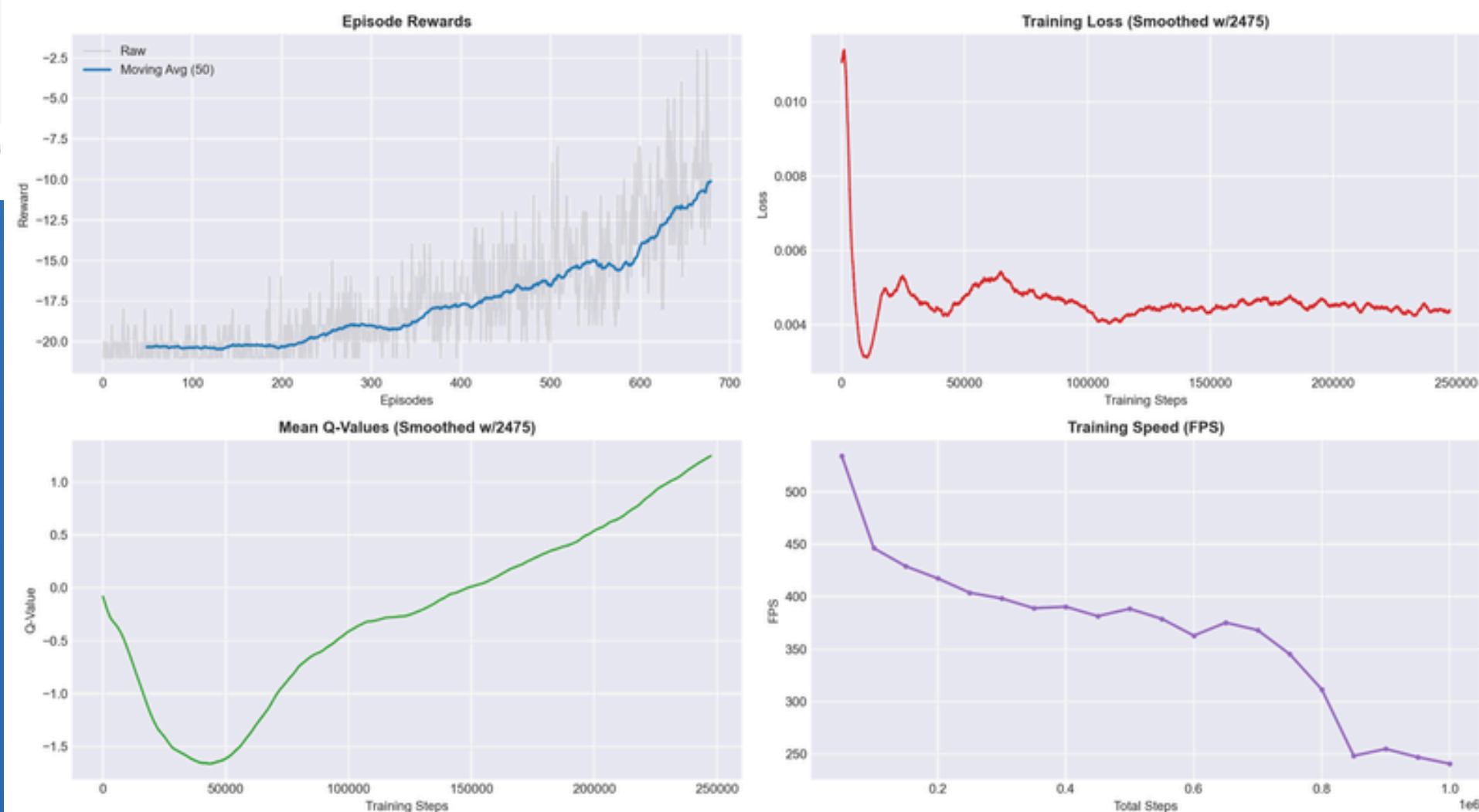


4. 5e-5, 64, 500_000,
200_000

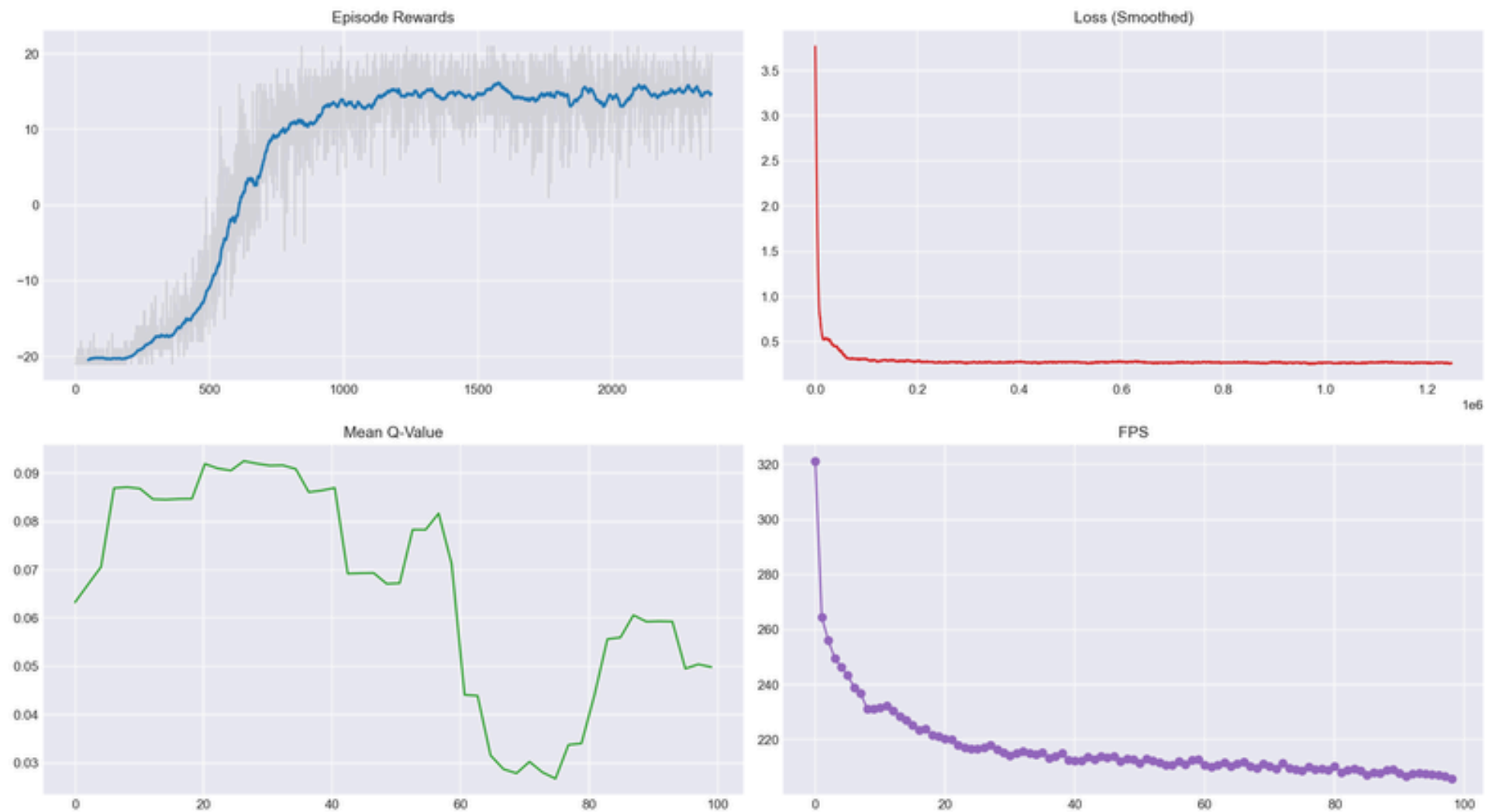
1.25e-4, 64, 1_000_000,
200_000

Pong - DQN

DQN Training Metrics (Decay=500000): PongNoFrameskip-v4

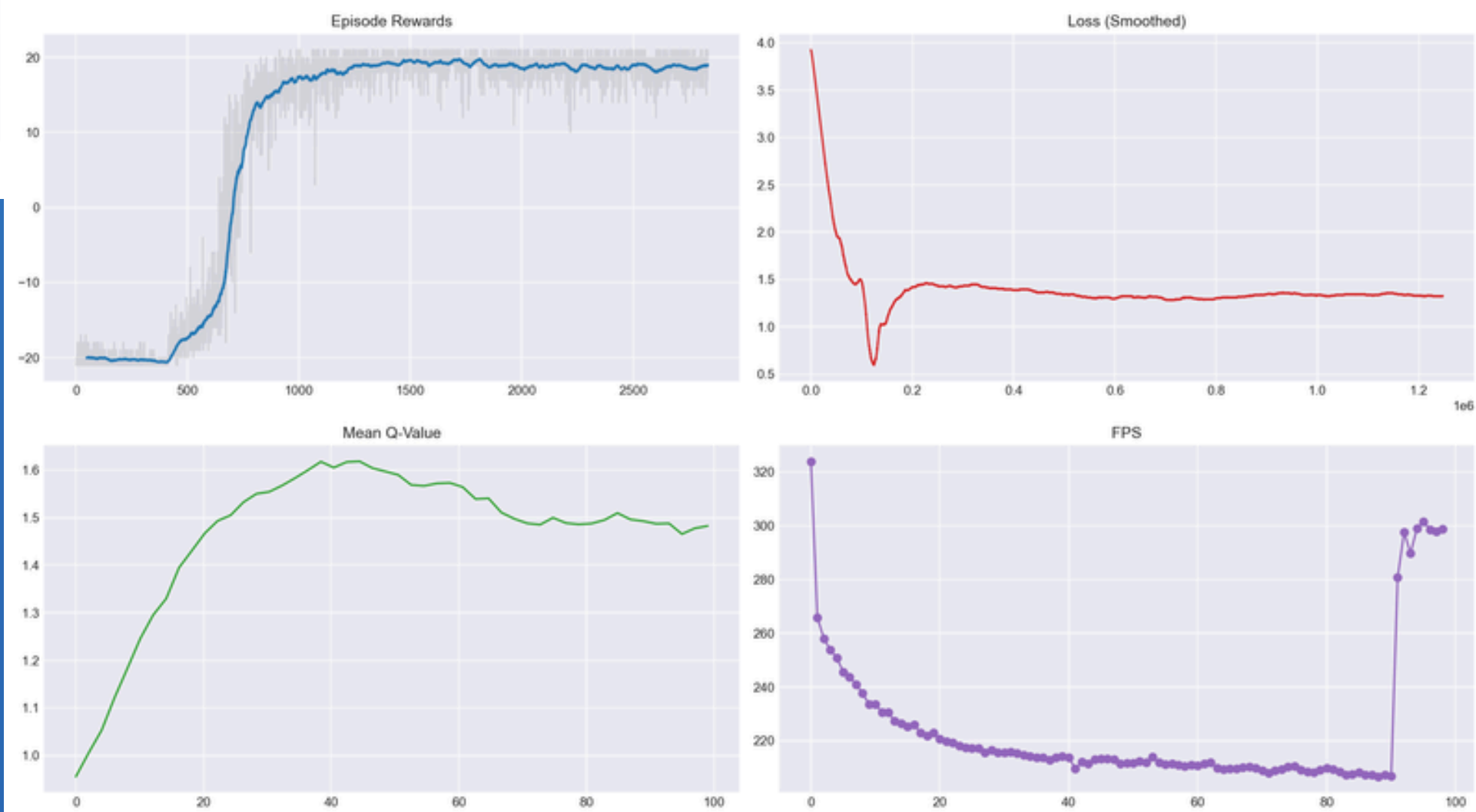


C51 Metrics - 3_Gamma_090



5. LR = 1e-4

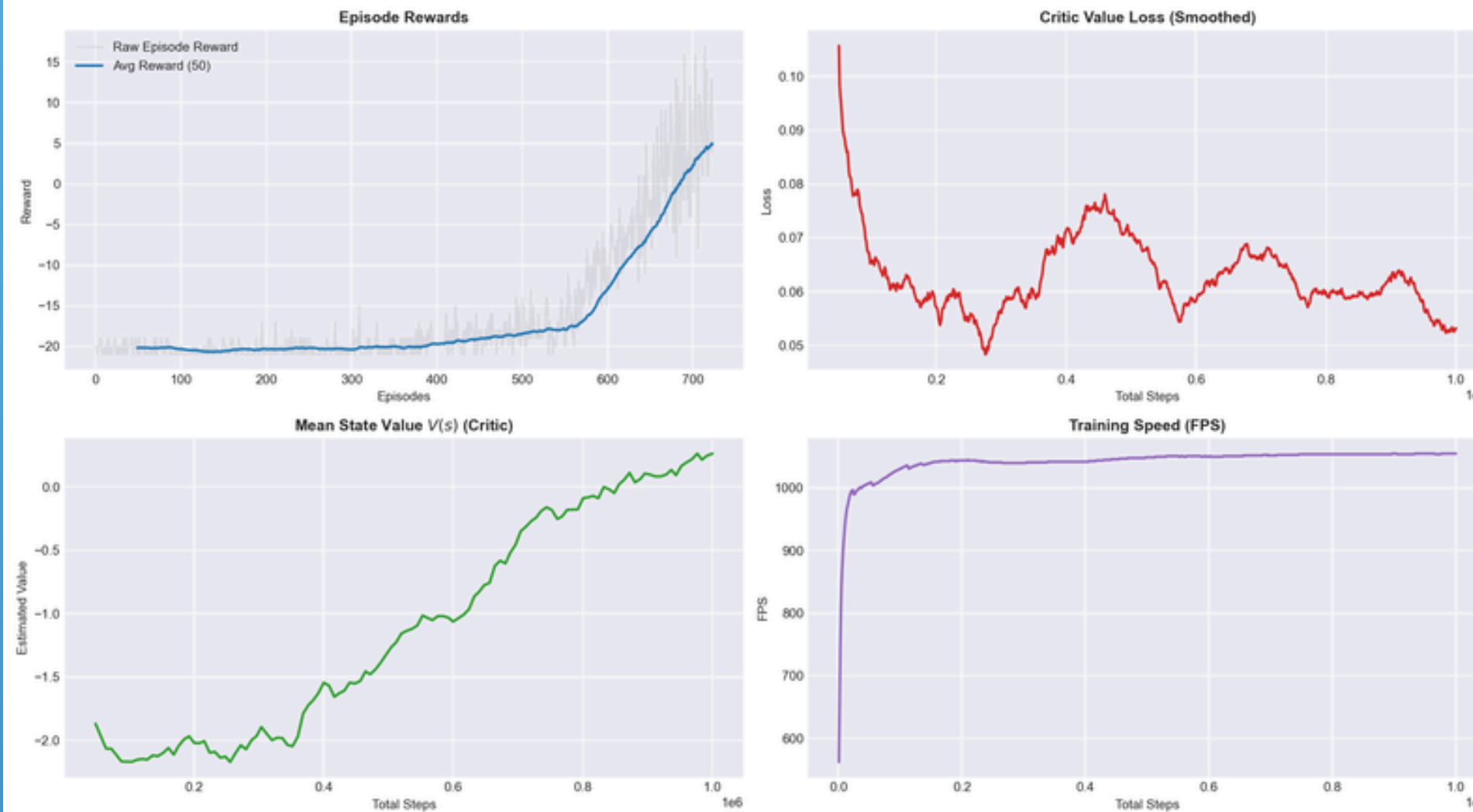
C51 Metrics - 5_LR_1e-4



"n_atoms": 51, "gamma": 0.90,
 "batch_size": 64, "lr": 2.5e-4 "buffer_size":
 100_000, "min_replay": 10_000,
 "target_tau": 0.005

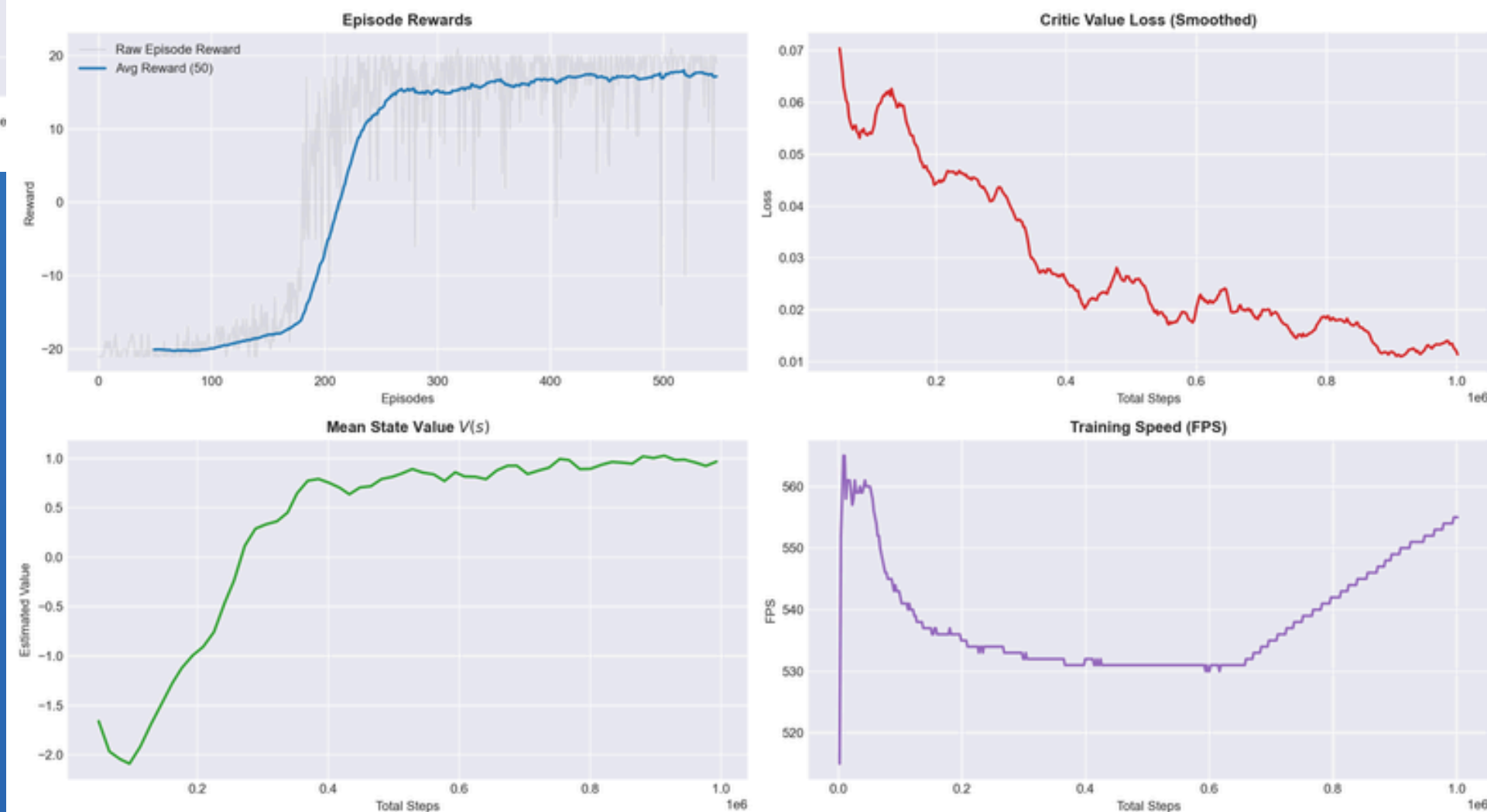
Pong - C51

PPO Training Metrics: PongNoFrameskip-v4



4. `lr_2.5e-4, _batch_size_64, _n_steps_128, _ent_coef_0.01`

PPO Metrics (LR=0.00025, BS=64): PongNoFrameskip-v4



1. `lr_2.5e-4, _batch_size_256, _n_steps_128, _ent_coef_0.01`

Pong - PPO

**Vă mulțumim
pentru
atenție!**

